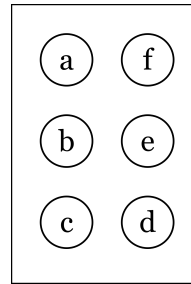


Function of one variable, group *Alpha* α -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 7 \arcsin\left(\frac{6-4x}{4}\right) - \frac{e^{5x+5}}{\sqrt{4x^2-6x}} \dots \dots \dots \left(\frac{3}{2}, \frac{5}{2}\right)$
 (b) $g(x) = \frac{1}{5} \ln(x^2 - 7x + 10) + \frac{\sqrt[3]{2-x}}{x-1} \dots \dots \dots (-\infty, 1) \cup (1, 2) \cup (5, \infty)$
 (c) $h(t) = \frac{\sqrt{16-t^2}}{\log_7(1+\frac{1}{3}t)} \dots \dots \dots (-3, 0) \cup (0, 4)$
 (d) $u(x) = \frac{-2x+5}{-4x-2} \dots \dots \dots (-\infty, \frac{-1}{2}) \cup (\frac{5}{2}, \infty)$
 (e) $\varphi(t) = -t^3 + 4t^2 - \frac{1}{t} - 2$, $t_0 = -\frac{1}{4} \dots \dots \dots -\frac{95}{64}$
 (f) $w(x) = -3x^3 - 8x^2 + 3x \dots \dots \dots 1$

1.



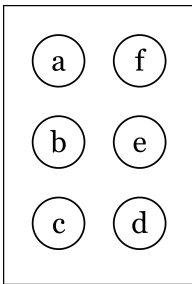
Braille letter

Function of one variable, group *Alpha* α -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -5 \arcsin\left(\frac{5-4x}{3}\right) - \frac{e^{-4x+7}}{\sqrt{12x^2-15x}} \dots \dots \dots \left(\frac{1}{2}, \frac{5}{4}\right)$
 (b) $g(x) = \frac{1}{3} \ln(x^2 - x - 30) + \frac{\sqrt[3]{-2-x}}{x-7} \dots \dots \dots (-\infty, -5) \cup (6, 7) \cup (7, \infty)$
 (c) $h(t) = \frac{\sqrt{81-t^2}}{\log_5(7+\frac{7}{8}t)} \dots \dots \dots (-8, -\frac{48}{7}) \cup (-\frac{48}{7}, 9)$
 (d) $u(x) = \frac{-2x+6}{4x+2} \dots \dots \dots (-\frac{1}{2}, 3)$
 (e) $\varphi(t) = -3t^3 + 3t^2 + \frac{3}{t} + 4$, $t_0 = -\frac{1}{3} \dots \dots \dots \frac{31}{9}$
 (f) $w(x) = \frac{x}{7} + \frac{7}{x} \dots \dots \dots 0$

2.



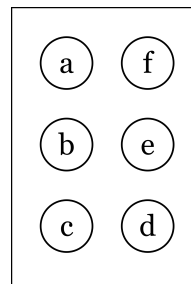
Braille letter

Function of one variable, group *Alpha* α -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -3 \arcsin\left(\frac{9-3x}{3}\right) - \frac{e^{-2x+4}}{\sqrt{3x^2-9x}} \dots \dots \dots (3, 4)$
 (b) $g(x) = \frac{1}{5} \ln(x^2 - 12x + 27) + \frac{\sqrt[3]{3-x}}{x-10} \dots \dots \dots (-\infty, 3) \cup (9, 10) \cup (10, \infty)$
 (c) $h(t) = \frac{\sqrt{4-t^2}}{\log_8(3-3t)} \dots \dots \dots \left(-2, \frac{2}{3}\right) \cup \left(\frac{2}{3}, 1\right)$
 (d) $u(x) = \frac{8x+5}{-5x-1} \dots \dots \dots \left(-\frac{5}{8}, \frac{-1}{5}\right)$
 (e) $\varphi(t) = -3t^3 + 5t^2 + \frac{2}{t} - 3$, $t_0 = -\frac{1}{2} \dots \dots \dots -\frac{43}{8}$
 (f) $w(x) = 6x^3 - 21x^2 - 45x \dots \dots \dots 2$

3.



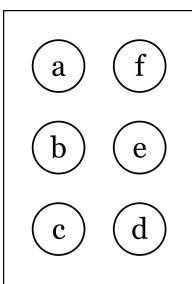
Braille letter

Function of one variable, group *Alpha* α -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -4 \arccos\left(\frac{4-2x}{2}\right) - \frac{e^{4x-3}}{\sqrt{-2x^2+4x}} \dots \dots \dots \langle 1, 2 \rangle$
 (b) $g(x) = \frac{1}{8} \ln(x^2 + 11x + 24) + \frac{\sqrt[3]{3-x}}{x+2} \dots \dots \dots (-\infty, -8) \cup (-3, -2) \cup (-2, \infty)$
 (c) $h(t) = \frac{\sqrt{49-t^2}}{\log_4(2+\frac{1}{3}t)} \dots \dots \dots (-6, -3) \cup (-3, 7)$
 (d) $u(x) = \frac{-2x-9}{6x+4} \dots \dots \dots \left(\frac{-9}{2}, \frac{-2}{3}\right)$
 (e) $\varphi(t) = 6t^3 + 5t^2 - \frac{5}{t} + 4$, $t_0 = -\frac{1}{2} \dots \dots \dots \frac{29}{2}$
 (f) $w(x) = \frac{x}{5} + \frac{5}{x} \dots \dots \dots 0$

4.



Braille letter

Function of one variable, group *Beta* β -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 4 \arcsin\left(\frac{6-5x}{4}\right) - \frac{e^{2x-6}}{\sqrt{10x^2-12x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{6}{5}, 2\right)$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 + 5x + 6) + \frac{\sqrt[3]{6-x}}{x+1} \quad ??? \quad (-\infty, -3) \cup (-2, -1) \cup (-1, \infty)$$

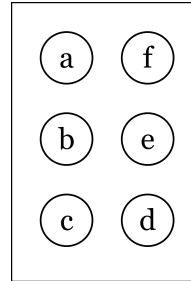
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_3(1-\frac{1}{3}t)} \quad \dots \quad ??? \quad \dots \quad \langle -4, 0 \rangle \cup (0, 3)$$

$$(d) \quad u(x) = \frac{2x-4}{3x-4} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{4}{3}) \cup (2, \infty)$$

$$(e) \quad \varphi(t) = 3t^3 + 8t^2 + \frac{2}{t} - 3, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{19}{8}$$

$$(f) \quad w(x) = \frac{x}{6} + \frac{6}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

1.



Braille letter

Function of one variable, group *Beta* β -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arcsin\left(\frac{3-x}{2}\right) - \frac{e^{-x+3}}{\sqrt{-x^2+3x}} \quad \dots \quad ??? \quad \dots \quad \langle 1, 3 \rangle$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 - 14x + 48) + \frac{\sqrt[3]{-4-x}}{x-5} \quad ??? \quad (-\infty, 5) \cup (5, 6) \cup (8, \infty)$$

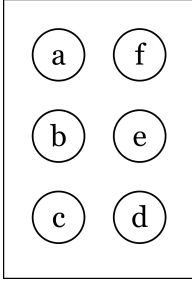
$$(c) \quad h(t) = \frac{\sqrt{49-t^2}}{\log_5(6-t)} \quad \dots \quad ??? \quad \dots \quad \langle -7, 5 \rangle \cup (5, 6)$$

$$(d) \quad u(x) = \frac{x+1}{5x+1} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup \left(\frac{-1}{5}, \infty\right)$$

$$(e) \quad \varphi(t) = 7t^3 - t^2 + \frac{8}{t} + 5, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{1}{8}$$

$$(f) \quad w(x) = -7x^3 - 9x^2 + 10x \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Beta* β -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arccos\left(\frac{6-3x}{2}\right) - \frac{e^{x-1}}{\sqrt{-24x^2+48x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{4}{3}, 2 \right\rangle$$

$$(b) \quad g(x) = \frac{1}{6} \ln(x^2 + 12x + 35) + \frac{\sqrt[3]{-1-x}}{x+8} \quad ??? \quad (-\infty, -8) \cup (-8, -7) \cup (-5, \infty)$$

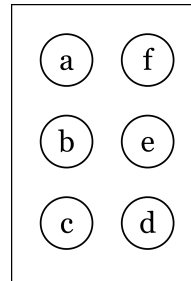
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_7(2-t)} \quad \dots \quad ??? \quad \dots \quad \langle -3, 1 \rangle \cup (1, 2)$$

$$(d) \quad u(x) = \frac{x+8}{-x+5} \quad \dots \quad ??? \quad \dots \quad \langle -8, 5 \rangle$$

$$(e) \quad \varphi(t) = 6t^3 - 3t^2 + \frac{4}{t} - 5, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{29}{2}$$

$$(f) \quad w(x) = \sqrt[3]{7x^2-x+1} \quad \dots \quad ??? \quad \dots \quad 1$$

3.



Braille letter

Function of one variable, group *Beta* β -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arcsin\left(\frac{3-x}{2}\right) - \frac{e^{7x+6}}{\sqrt{-4x^2+12x}} \quad \dots \quad ??? \quad \dots \quad \langle 1, 3 \rangle$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + x - 6) + \frac{\sqrt[3]{-3-x}}{x-3} \quad ??? \quad (-\infty, -3) \cup (2, 3) \cup (3, \infty)$$

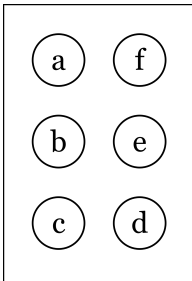
$$(c) \quad h(t) = \frac{\sqrt{25-t^2}}{\log_3(8-2t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -5, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 4\right)$$

$$(d) \quad u(x) = \frac{5x+2}{4x+1} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{-2}{5}) \cup \left(\frac{-1}{4}, \infty\right)$$

$$(e) \quad \varphi(t) = 2t^3 + 7t^2 + \frac{3}{t} + 7, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{181}{27}$$

$$(f) \quad w(x) = \sqrt[5]{-4x^2+x-9} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



Braille letter

Function of one variable, group *Gamma* γ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arcsin\left(\frac{8-x}{5}\right) - \frac{e^{-9x+2}}{\sqrt{2x^2-16x}} \quad \dots \quad ??? \quad \dots \quad \left\langle 3, 8 \right\rangle$$

$$(b) \quad g(x) = \frac{1}{7} \ln(x^2 - 3x - 4) + \frac{\sqrt[3]{8-x}}{x-5} \quad ??? \quad (-\infty, -1) \cup (4, 5) \cup (5, \infty)$$

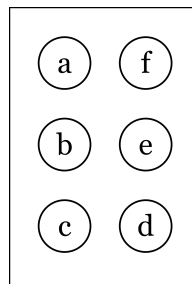
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_5(2+2t)} \quad \dots \quad ??? \quad \dots \quad \left(-1, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 2\right)$$

$$(d) \quad u(x) = \frac{x+2}{5x-1} \quad \dots \quad ??? \quad \dots \quad (-\infty, -2) \cup \left(\frac{1}{5}, \infty\right)$$

$$(e) \quad \varphi(t) = -5t^3 - t^2 + \frac{5}{t} + 2, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{61}{8}$$

$$(f) \quad w(x) = \sqrt[5]{-8x^2 - 3x - 2} \quad \dots \quad ??? \quad \dots \quad 0$$

1.



Braille letter

Function of one variable, group *Gamma* γ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arccos\left(\frac{9-4x}{2}\right) - \frac{e^{-4x+1}}{\sqrt{-16x^2+36x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{7}{4}, \frac{9}{4} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{8} \ln(x^2 - 11x + 30) + \frac{\sqrt[3]{6-x}}{x-4} \quad ??? \quad (-\infty, 4) \cup (4, 5) \cup (5, \infty)$$

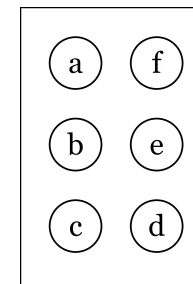
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_2(2+t)} \quad \dots \quad ??? \quad \dots \quad (-2, -1) \cup (-1, 3)$$

$$(d) \quad u(x) = \frac{-9x-3}{x+6} \quad \dots \quad ??? \quad \dots \quad \left(-6, -\frac{1}{3}\right)$$

$$(e) \quad \varphi(t) = -3t^3 + t^2 - \frac{2}{t} - 8, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{27}{8}$$

$$(f) \quad w(x) = \frac{x}{2} + \frac{2}{x} \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Gamma* γ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -4 \arccos\left(\frac{8-2x}{3}\right) - \frac{e^{-2x-5}}{\sqrt{4x^2-16x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{5}{2}, 4 \right\rangle$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 + 12x + 35) + \frac{\sqrt[3]{3-x}}{x+4} \quad ??? \quad (-\infty, -7) \cup (-5, -4) \cup (-4, \infty)$$

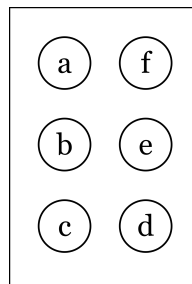
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_2(1-t)} \quad \dots \quad ??? \quad \dots \quad \langle -2, 0 \rangle \cup (0, 1)$$

$$(d) \quad u(x) = \frac{2x-1}{-7x-4} \quad \dots \quad ??? \quad \dots \quad \left(-\frac{4}{7}, \frac{1}{2}\right)$$

$$(e) \quad \varphi(t) = -9t^3 + 7t^2 + \frac{7}{t} - 8, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{83}{9}$$

$$(f) \quad w(x) = \sqrt[3]{8x^2 - 5x + 6} \quad \dots \quad ??? \quad \dots \quad 0$$

3.



Braille letter

Function of one variable, group *Gamma* γ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arccos\left(\frac{4-6x}{2}\right) - \frac{e^{x+2}}{\sqrt{-6x^2+4x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{2}{3}, 1\right)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 - 2x - 15) + \frac{\sqrt[3]{-5-x}}{x-6} \quad ??? \quad (-\infty, -3) \cup (5, 6) \cup (6, \infty)$$

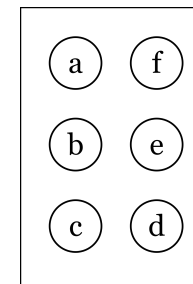
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_4(7-7t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -2, \frac{6}{7} \right\rangle \cup \left(\frac{6}{7}, 1\right)$$

$$(d) \quad u(x) = \frac{-4x-3}{-x-4} \quad \dots \quad ??? \quad \dots \quad (-\infty, -4) \cup \left(-\frac{3}{4}, \infty\right)$$

$$(e) \quad \varphi(t) = 4t^3 - 8t^2 + \frac{6}{t} - 6, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{489}{16}$$

$$(f) \quad w(x) = \sqrt[5]{-5x^2 + 4x - 7} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



Braille letter

Function of one variable, group *Delta* δ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -8 \arcsin\left(\frac{9-3x}{4}\right) - \frac{e^{8x+6}}{\sqrt{15x^2-45x}} \quad \dots \quad ??? \quad \dots \quad \left(3, \frac{13}{3}\right)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 5x + 4) + \frac{\sqrt[3]{1-x}}{x-5} \quad ??? \quad (-\infty, 1) \cup (4, 5) \cup (5, \infty)$$

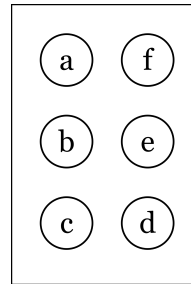
$$(c) \quad h(t) = \frac{\sqrt{49-t^2}}{\log_4(1+\frac{1}{6}t)} \quad \dots \quad ??? \quad \dots \quad (-6, 0) \cup (0, 7)$$

$$(d) \quad u(x) = \frac{-2x+6}{-7x+3} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{3}{7}) \cup (3, \infty)$$

$$(e) \quad \varphi(t) = -2t^3 + t^2 - \frac{5}{t} + 4, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad 7$$

$$(f) \quad w(x) = \sqrt[7]{x^2-2x+4} \quad \dots \quad ??? \quad \dots \quad 1$$

1.



Braille letter

Function of one variable, group *Delta* δ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = \arcsin\left(\frac{8-5x}{4}\right) - \frac{e^{3x-7}}{\sqrt{-10x^2+16x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{4}{5}, \frac{8}{5} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{7} \ln(x^2 - 2x - 35) + \frac{\sqrt[3]{1-x}}{x-8} \quad ??? \quad (-\infty, -5) \cup (7, 8) \cup (8, \infty)$$

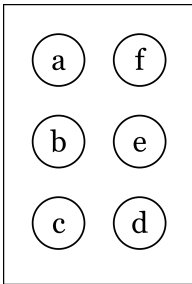
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_9(2+\frac{2}{3}t)} \quad \dots \quad ??? \quad \dots \quad (-3, -\frac{3}{2}) \cup (-\frac{3}{2}, 4)$$

$$(d) \quad u(x) = \frac{-x-4}{-4x-4} \quad \dots \quad ??? \quad \dots \quad (-\infty, -4) \cup (-1, \infty)$$

$$(e) \quad \varphi(t) = 2t^3 - 5t^2 - \frac{4}{t} - 6, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad \frac{1}{2}$$

$$(f) \quad w(x) = \sqrt[7]{-7x^2-x-5} \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Delta* δ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = \arccos\left(\frac{6-2x}{2}\right) - \frac{e^{8x+1}}{\sqrt{-6x^2+18x}} \quad \dots \quad ??? \quad \dots \quad (2, 3)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 + 7x + 12) + \frac{\sqrt[3]{1-x}}{x+5} \quad ??? \quad (-\infty, -5) \cup (-5, -4) \cup (-3, \infty)$$

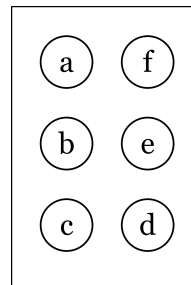
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_6(2+2t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{1}{2}) \cup (-\frac{1}{2}, 2)$$

$$(d) \quad u(x) = \frac{4x+7}{2x-2} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{-7}{4}) \cup (1, \infty)$$

$$(e) \quad \varphi(t) = -5t^3 - t^2 - \frac{3}{t} + 4, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad \frac{305}{64}$$

$$(f) \quad w(x) = x^3 - 7x^2 - 8x \quad \dots \quad ??? \quad \dots \quad 3$$

3.



Braille letter

Function of one variable, group *Delta* δ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -8 \arcsin\left(\frac{5-4x}{2}\right) - \frac{e^{-9x+2}}{\sqrt{-8x^2+10x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{3}{4}, \frac{5}{4} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 + 6x + 8) + \frac{\sqrt[3]{5-x}}{x+5} \quad ??? \quad (-\infty, -5) \cup (-5, -4) \cup (-2, \infty)$$

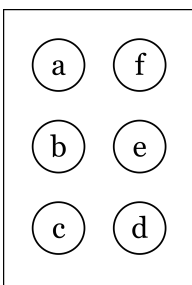
$$(c) \quad h(t) = \frac{\sqrt{81-t^2}}{\log_6(5+\frac{5}{8}t)} \quad \dots \quad ??? \quad \dots \quad (-8, -\frac{32}{5}) \cup (-\frac{32}{5}, 9)$$

$$(d) \quad u(x) = \frac{-6x+4}{5x-5} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{2}{3}, 1 \right\rangle$$

$$(e) \quad \varphi(t) = 2t^3 - 3t^2 - \frac{3}{t} - 5, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad \frac{217}{32}$$

$$(f) \quad w(x) = \frac{x}{4} + \frac{4}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



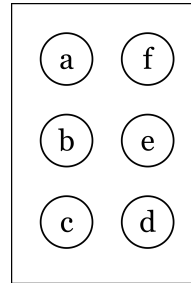
Braille letter

Function of one variable, group *Epsilon* ϵ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -4 \arcsin\left(\frac{3-5x}{2}\right) - \frac{e^{4x-8}}{\sqrt{-35x^2+21x}}$... ??? ... $\left\langle \frac{1}{5}, \frac{3}{5} \right\rangle$
 (b) $g(x) = \frac{1}{5} \ln(x^2 - 5x - 24) + \frac{\sqrt[3]{-6-x}}{x-9}$??? $(-\infty, -3) \cup (8, 9) \cup (9, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_5(3+\frac{3}{5}t)}$??? $(-5, -\frac{10}{3}) \cup (-\frac{10}{3}, 6)$
 (d) $u(x) = \frac{-5x+3}{-5x+2}$??? $(-\infty, \frac{2}{5}) \cup (\frac{3}{5}, \infty)$
 (e) $\varphi(t) = t^3 + 9t^2 + \frac{6}{t} + 1$, $t_0 = -\frac{1}{2}$??? $-\frac{71}{8}$
 (f) $w(x) = \sqrt[7]{-2x^2-3x-4}$??? 1

1.



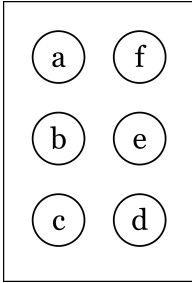
Braille letter

Function of one variable, group *Epsilon* ϵ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 6 \arccos\left(\frac{9-7x}{4}\right) - \frac{e^{-6x-6}}{\sqrt{21x^2-27x}}$... ??? ... $\left(\frac{9}{7}, \frac{13}{7}\right)$
 (b) $g(x) = \frac{1}{3} \ln(x^2 + 2x - 8) + \frac{\sqrt[3]{4-x}}{x-3}$??? $(-\infty, -4) \cup (2, 3) \cup (3, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_2(2-t)}$??? $\langle -3, 1) \cup (1, 2)$
 (d) $u(x) = \frac{-x-6}{-3x+2}$??? $(-\infty, -6) \cup (\frac{2}{3}, \infty)$
 (e) $\varphi(t) = -5t^3 - 3t^2 + \frac{9}{t} + 3$, $t_0 = -\frac{1}{2}$??? $-\frac{13}{8}$
 (f) $w(x) = \sqrt[3]{8x^2-3x+4}$??? 1

2.



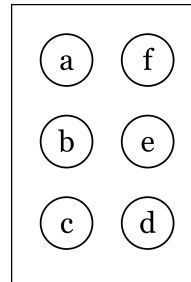
Braille letter

Function of one variable, group *Epsilon* ϵ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 3 \arccos\left(\frac{4-x}{2}\right) - \frac{e^{-5x+3}}{\sqrt{7x^2-28x}}$??? $\langle 2, 4)$
 (b) $g(x) = \frac{1}{4} \ln(x^2 - 6x - 16) + \frac{\sqrt[3]{-4-x}}{x-9}$??? $(-\infty, -2) \cup (8, 9) \cup (9, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_2(5+\frac{5}{2}t)}$??? $(-2, -\frac{8}{5}) \cup (-\frac{8}{5}, 3)$
 (d) $u(x) = \frac{-8x-7}{9x+1}$??? $(-\frac{7}{8}, \frac{-1}{9})$
 (e) $\varphi(t) = -2t^3 + 3t^2 + \frac{4}{t} + 7$, $t_0 = -\frac{1}{3}$??? $\frac{164}{27}$
 (f) $w(x) = \sqrt[7]{5x^2-5x+6}$??? 0

3.



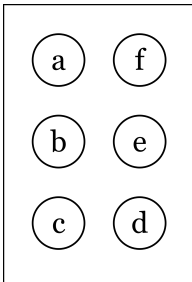
Braille letter

Function of one variable, group *Epsilon* ϵ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = \arcsin\left(\frac{9-6x}{5}\right) - \frac{e^{-2x-5}}{\sqrt{-24x^2+36x}}$??? $(\frac{3}{2}, \frac{7}{3})$
 (b) $g(x) = \frac{1}{3} \ln(x^2 + 3x - 54) + \frac{\sqrt[3]{-2-x}}{x+10}$??? $(-\infty, -10) \cup (-10, -9) \cup (6, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_9(1+\frac{1}{2}t)}$??? $(-2, 0) \cup (0, 3)$
 (d) $u(x) = \frac{3x+3}{4x-2}$??? $(-\infty, -1) \cup (\frac{1}{2}, \infty)$
 (e) $\varphi(t) = 2t^3 - 7t^2 + \frac{9}{t} - 4$, $t_0 = -\frac{1}{3}$??? $-\frac{860}{27}$
 (f) $w(x) = \sqrt[5]{8x^2+4x+5}$??? 0

4.



Braille letter

Function of one variable, group *Zeta* ζ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -4 \arccos\left(\frac{3-8x}{2}\right) - \frac{e^{8x-4}}{\sqrt{24x^2-9x}} \dots \dots \dots \left\langle \frac{1}{8}, \frac{3}{8} \right\rangle$

(b) $g(x) = \frac{1}{7} \ln(x^2 - 2x - 35) + \frac{\sqrt[3]{4-x}}{x-8} \dots \dots \dots (-\infty, -5) \cup (7, 8) \cup (8, \infty)$

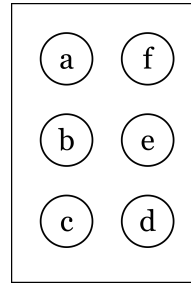
(c) $h(t) = \frac{\sqrt{9-t^2}}{\log_4(3-\frac{3}{2}t)} \dots \dots \dots \left\langle -3, \frac{4}{3} \right\rangle \cup \left(\frac{4}{3}, 2 \right)$

(d) $u(x) = \frac{-2x+7}{4x-6} \dots \dots \dots \left\langle \frac{3}{2}, \frac{7}{2} \right\rangle$

(e) $\varphi(t) = -5t^3 + 3t^2 - \frac{7}{t} + 7$, $t_0 = -\frac{1}{3} \dots \dots \dots \frac{770}{27}$

(f) $w(x) = x^3 + x^2 - 6x \dots \dots \dots 3$

1.



Braille letter

Function of one variable, group *Zeta* ζ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 5 \arcsin\left(\frac{9-x}{4}\right) - \frac{e^{5x-9}}{\sqrt{2x^2-18x}} \dots \dots \dots (9, 13)$

(b) $g(x) = \frac{1}{7} \ln(x^2 + 16x + 63) + \frac{\sqrt[3]{6-x}}{x+6} \dots \dots \dots (-\infty, -9) \cup (-7, -6) \cup (-6, \infty)$

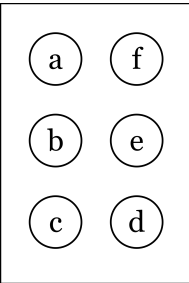
(c) $h(t) = \frac{\sqrt{25-t^2}}{\log_4(1+\frac{1}{4}t)} \dots \dots \dots (-4, 0) \cup (0, 5)$

(d) $u(x) = \frac{-5x+4}{-4x-2} \dots \dots \dots \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{4}{5}, \infty\right)$

(e) $\varphi(t) = 5t^3 + 5t^2 + \frac{4}{t} + 3$, $t_0 = -\frac{1}{2} \dots \dots \dots -\frac{35}{8}$

(f) $w(x) = \sqrt[3]{3x^2+3x+3} \dots \dots \dots 1$

2.



Braille letter

Function of one variable, group *Zeta* ζ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 4 \arccos\left(\frac{7-5x}{3}\right) - \frac{e^{-7x+2}}{\sqrt{30x^2-42x}} \dots \dots \dots \left\langle \frac{7}{5}, 2 \right\rangle$

(b) $g(x) = \frac{1}{7} \ln(x^2 + x - 6) + \frac{\sqrt[3]{-7-x}}{x-3} \dots \dots \dots (-\infty, -3) \cup (2, 3) \cup (3, \infty)$

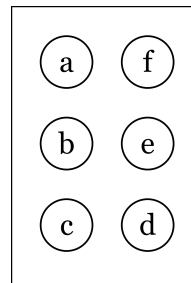
(c) $h(t) = \frac{\sqrt{9-t^2}}{\log_9(2+t)} \dots \dots \dots (-2, -1) \cup (-1, 3)$

(d) $u(x) = \frac{-6x+2}{6x+1} \dots \dots \dots \left(-\frac{1}{6}, \frac{1}{3}\right)$

(e) $\varphi(t) = 4t^3 + 2t^2 + \frac{5}{t} + 8$, $t_0 = -\frac{1}{3} \dots \dots \dots \frac{173}{27}$

(f) $w(x) = 4x^3 + 26x^2 + 30x \dots \dots \dots 1$

3.



Braille letter

Function of one variable, group *Zeta* ζ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 2 \arcsin\left(\frac{6-8x}{5}\right) - \frac{e^{5x-2}}{\sqrt{-56x^2+42x}} \dots \dots \dots \left\langle \frac{1}{8}, \frac{3}{4} \right\rangle$

(b) $g(x) = \frac{1}{2} \ln(x^2 - 9x + 20) + \frac{\sqrt[3]{1-x}}{x-3} \dots \dots \dots (-\infty, 3) \cup (3, 4) \cup (5, \infty)$

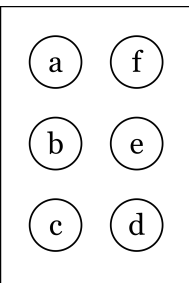
(c) $h(t) = \frac{\sqrt{4-t^2}}{\log_6(5+5t)} \dots \dots \dots \left(-1, -\frac{4}{5}\right) \cup \left(-\frac{4}{5}, 2\right)$

(d) $u(x) = \frac{-x+4}{-3x+7} \dots \dots \dots \left(-\infty, \frac{7}{3}\right) \cup (4, \infty)$

(e) $\varphi(t) = 5t^3 - 2t^2 - \frac{3}{t} - 3$, $t_0 = -\frac{1}{3} \dots \dots \dots -\frac{65}{27}$

(f) $w(x) = -3x^3 - 9x^2 + 54x \dots \dots \dots 3$

4.



Braille letter

Function of one variable, group *Eta* η -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 3 \arccos\left(\frac{9-x}{3}\right) - \frac{e^{4x+6}}{\sqrt{-5x^2+45x}} \quad \dots \quad ??? \quad \dots \quad (9, 12)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + 5x + 4) + \frac{\sqrt[3]{-4-x}}{x+5} \quad ??? \quad (-\infty, -5) \cup (-5, -4) \cup (-1, \infty)$$

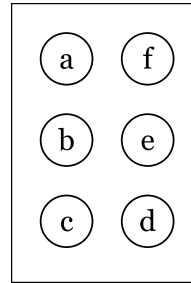
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_8(7+\frac{7}{2}t)} \quad \dots \quad ??? \quad \dots \quad (-2, -\frac{12}{7}) \cup (-\frac{12}{7}, 3)$$

$$(d) \quad u(x) = \frac{3x-6}{-2x+5} \quad \dots \quad ??? \quad \dots \quad (2, \frac{5}{2})$$

$$(e) \quad \varphi(t) = -2t^3 + t^2 + \frac{4}{t} + 4, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad \frac{5}{2}$$

$$(f) \quad w(x) = 8x^3 + 46x^2 + 30x \quad \dots \quad ??? \quad \dots \quad 3$$

1.



Braille letter

Function of one variable, group *Eta* η -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -9 \arcsin\left(\frac{6-6x}{3}\right) - \frac{e^{5x+4}}{\sqrt{6x^2-6x}} \quad \dots \quad ??? \quad \dots \quad (1, \frac{3}{2})$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 + 6x - 16) + \frac{\sqrt[3]{-1-x}}{x-3} \quad ??? \quad (-\infty, -8) \cup (2, 3) \cup (3, \infty)$$

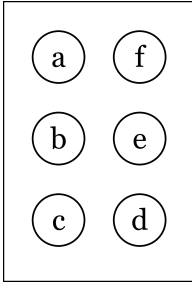
$$(c) \quad h(t) = \frac{\sqrt{36-t^2}}{\log_7(5+t)} \quad \dots \quad ??? \quad \dots \quad (-5, -4) \cup (-4, 6)$$

$$(d) \quad u(x) = \frac{2x+1}{5x-1} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{-1}{2}) \cup (\frac{1}{5}, \infty)$$

$$(e) \quad \varphi(t) = -6t^3 - 9t^2 + \frac{2}{t} + 4, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{143}{32}$$

$$(f) \quad w(x) = -\frac{x}{5} - \frac{5}{x} \quad \dots \quad ??? \quad \dots \quad 1$$

2.



Braille letter

Function of one variable, group *Eta* η -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arccos\left(\frac{9-3x}{5}\right) - \frac{e^{-7x-2}}{\sqrt{-3x^2+9x}} \quad \dots \quad ??? \quad \dots \quad \langle \frac{4}{3}, 3 \rangle$$

$$(b) \quad g(x) = \frac{1}{6} \ln(x^2 - x - 6) + \frac{\sqrt[3]{-4-x}}{x-4} \quad ??? \quad (-\infty, -2) \cup (3, 4) \cup (4, \infty)$$

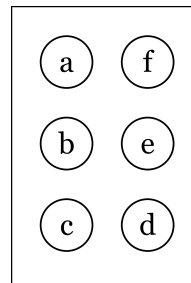
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_3(6-2t)} \quad \dots \quad ??? \quad \dots \quad \langle -4, \frac{5}{2} \rangle \cup (\frac{5}{2}, 3)$$

$$(d) \quad u(x) = \frac{-7x-7}{-7x+1} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup (\frac{1}{7}, \infty)$$

$$(e) \quad \varphi(t) = -5t^3 + 7t^2 - \frac{5}{t} - 6, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad \frac{929}{64}$$

$$(f) \quad w(x) = -x^3 + 4x^2 + 5x \quad \dots \quad ??? \quad \dots \quad 2$$

3.



Braille letter

Function of one variable, group *Eta* η -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arcsin\left(\frac{4-6x}{2}\right) - \frac{e^{7x-6}}{\sqrt{-6x^2+4x}} \quad \dots \quad ??? \quad \dots \quad \langle \frac{1}{3}, \frac{2}{3} \rangle$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 - 9x + 18) + \frac{\sqrt[3]{4-x}}{x-7} \quad ??? \quad (-\infty, 3) \cup (6, 7) \cup (7, \infty)$$

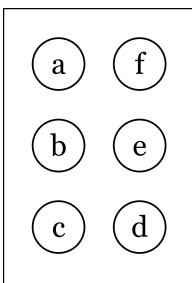
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_5(1+\frac{1}{2}t)} \quad \dots \quad ??? \quad \dots \quad (-2, 0) \cup (0, 3)$$

$$(d) \quad u(x) = \frac{-4x-4}{-2x+5} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup (\frac{5}{2}, \infty)$$

$$(e) \quad \varphi(t) = -5t^3 - 6t^2 - \frac{4}{t} + 2, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{77}{27}$$

$$(f) \quad w(x) = -9x^3 + 15x^2 + 6x \quad \dots \quad ??? \quad \dots \quad 3$$

4.



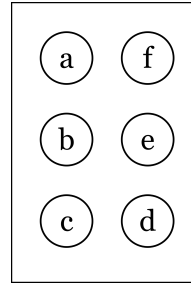
Braille letter

Function of one variable, group *Theta* θ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -\arcsin\left(\frac{6-2x}{2}\right) - \frac{e^{4x+4}}{\sqrt{-4x^2+12x}} \dots ??? \dots \langle 2, 3 \rangle$
 (b) $g(x) = \frac{1}{2} \ln(x^2 - 10x + 21) + \frac{\sqrt[3]{1-x}}{x-8} ??? \dots (-\infty, 3) \cup (7, 8) \cup (8, \infty)$
 (c) $h(t) = \frac{\sqrt{4-t^2}}{\log_2(2+2t)} \dots ??? \dots \dots (-1, -\frac{1}{2}) \cup (-\frac{1}{2}, 2)$
 (d) $u(x) = \frac{4x-1}{4x-5} \dots ??? \dots \dots (-\infty, \frac{1}{4}) \cup (\frac{5}{4}, \infty)$
 (e) $\varphi(t) = -6t^3 + 4t^2 + \frac{6}{t} - 4, t_0 = -\frac{1}{2} \dots ??? \dots -\frac{21}{4}$
 (f) $w(x) = \sqrt[7]{6x^2+4x+4} \dots ??? \dots \dots 2$

1.



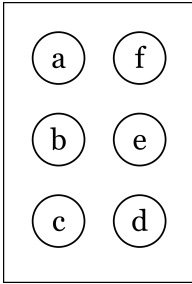
Braille letter

Function of one variable, group *Theta* θ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -2 \arcsin\left(\frac{5-5x}{4}\right) - \frac{e^{9x+7}}{\sqrt{-20x^2+20x}} \dots ??? \dots \langle \frac{1}{5}, 1 \rangle$
 (b) $g(x) = \frac{1}{4} \ln(x^2 + x - 30) + \frac{\sqrt[3]{8-x}}{x+7} ??? \dots (-\infty, -7) \cup (-7, -6) \cup (5, \infty)$
 (c) $h(t) = \frac{\sqrt{4-t^2}}{\log_6(7-7t)} \dots ??? \dots \dots \langle -2, \frac{6}{7} \rangle \cup (\frac{6}{7}, 1)$
 (d) $u(x) = \frac{-3x+5}{-5x+6} \dots ??? \dots \dots (-\infty, \frac{6}{5}) \cup (\frac{5}{3}, \infty)$
 (e) $\varphi(t) = 8t^3 - 3t^2 + \frac{3}{t} - 6, t_0 = -\frac{1}{4} \dots ??? \dots -\frac{113}{16}$
 (f) $w(x) = -\frac{x}{6} - \frac{6}{x} \dots ??? \dots \dots 1$

2.



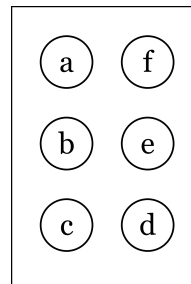
Braille letter

Function of one variable, group *Theta* θ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 7 \arccos\left(\frac{7-3x}{5}\right) - \frac{e^{8x-4}}{\sqrt{6x^2-14x}} \dots ??? \dots \dots (\frac{7}{3}, 4)$
 (b) $g(x) = \frac{1}{5} \ln(x^2 - 12x + 27) + \frac{\sqrt[3]{1-x}}{x-2} ??? \dots (-\infty, 2) \cup (2, 3) \cup (9, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_3(1+\frac{1}{2}t)} \dots ??? \dots \dots (-2, 0) \cup (0, 3)$
 (d) $u(x) = \frac{-3x-6}{3x+1} \dots ??? \dots \dots (-2, \frac{-1}{3})$
 (e) $\varphi(t) = 5t^3 + 2t^2 + \frac{5}{t} + 3, t_0 = -\frac{1}{2} \dots ??? \dots -\frac{57}{8}$
 (f) $w(x) = \sqrt[3]{-8x^2+9x-8} \dots ??? \dots \dots 0$

3.



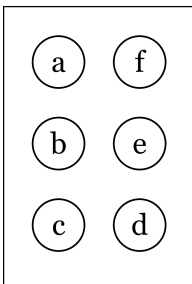
Braille letter

Function of one variable, group *Theta* θ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 4 \arcsin\left(\frac{6-5x}{2}\right) - \frac{e^{3x-5}}{\sqrt{10x^2-12x}} \dots ??? \dots \dots (\frac{6}{5}, \frac{8}{5})$
 (b) $g(x) = \frac{1}{4} \ln(x^2 - x - 12) + \frac{\sqrt[3]{2-x}}{x+4} ??? \dots (-\infty, -4) \cup (-4, -3) \cup (4, \infty)$
 (c) $h(t) = \frac{\sqrt{16-t^2}}{\log_4(7-\frac{7}{3}t)} \dots ??? \dots \dots \langle -4, \frac{18}{7} \rangle \cup (\frac{18}{7}, 3)$
 (d) $u(x) = \frac{2x-7}{7x-1} \dots ??? \dots \dots (-\infty, \frac{1}{7}) \cup (\frac{7}{2}, \infty)$
 (e) $\varphi(t) = 2t^3 + t^2 + \frac{4}{t} + 2, t_0 = -\frac{1}{3} \dots ??? \dots -\frac{269}{27}$
 (f) $w(x) = -3x^3 - 9x^2 - 6x \dots ??? \dots \dots 3$

4.



Braille letter

Function of one variable, group *Iota* ι -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 2 \arcsin\left(\frac{6-9x}{5}\right) - \frac{e^{3x+1}}{\sqrt{-81x^2 + 54x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{1}{9}, \frac{2}{3} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 7x + 10) + \frac{\sqrt[3]{7-x}}{x-1} \quad ??? \quad (-\infty, 1) \cup (1, 2) \cup (5, \infty)$$

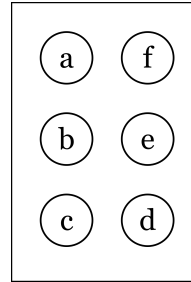
$$(c) \quad h(t) = \frac{\sqrt{25-t^2}}{\log_3(4-t)} \quad \dots \quad ??? \quad \dots \quad (-5, 3) \cup (3, 4)$$

$$(d) \quad u(x) = \frac{4x+5}{-3x+2} \quad \dots \quad ??? \quad \dots \quad \left\langle -\frac{5}{4}, \frac{2}{3} \right\rangle$$

$$(e) \quad \varphi(t) = 2t^3 - 2t^2 + \frac{5}{t} - 2, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{107}{27}$$

$$(f) \quad w(x) = 4x^3 + 34x^2 + 42x \quad \dots \quad ??? \quad \dots \quad 3$$

1.



Braille letter

Function of one variable, group *Iota* ι -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arccos\left(\frac{7-3x}{4}\right) - \frac{e^{5x-3}}{\sqrt{18x^2 - 42x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{7}{3}, \frac{11}{3} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 9x + 20) + \frac{\sqrt[3]{-2-x}}{x-6} \quad ??? \quad (-\infty, 4) \cup (5, 6) \cup (6, \infty)$$

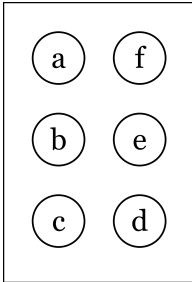
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_3(9+3t)} \quad \dots \quad ??? \quad \dots \quad (-3, -\frac{8}{3}) \cup (-\frac{8}{3}, 4)$$

$$(d) \quad u(x) = \frac{6x-1}{-4x-4} \quad \dots \quad ??? \quad \dots \quad (-1, \frac{1}{6})$$

$$(e) \quad \varphi(t) = 6t^3 - 6t^2 + \frac{7}{t} - 4, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{1039}{32}$$

$$(f) \quad w(x) = -\frac{x}{7} - \frac{7}{x} \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Iota* ι -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arcsin\left(\frac{8-3x}{4}\right) - \frac{e^{-6x+7}}{\sqrt{-15x^2 + 40x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{4}{3}, \frac{8}{3} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{6} \ln(x^2 + 4x - 5) + \frac{\sqrt[3]{7-x}}{x+6} \quad ??? \quad (-\infty, -6) \cup (-6, -5) \cup (1, \infty)$$

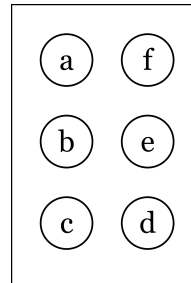
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_8(4+4t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{3}{4}) \cup (-\frac{3}{4}, 2)$$

$$(d) \quad u(x) = \frac{5x+4}{6x+3} \quad \dots \quad ??? \quad \dots \quad (-\infty, -\frac{4}{5}) \cup (-\frac{1}{2}, \infty)$$

$$(e) \quad \varphi(t) = -3t^3 + 3t^2 + \frac{5}{t} - 2, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{87}{8}$$

$$(f) \quad w(x) = -9x^3 - 12x^2 - 3x \quad \dots \quad ??? \quad \dots \quad 2$$

3.



Braille letter

Function of one variable, group *Iota* ι -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 4 \arccos\left(\frac{8-x}{2}\right) - \frac{e^{-6x-3}}{\sqrt{5x^2 - 40x}} \quad \dots \quad ??? \quad \dots \quad (6, 8)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 + 5x - 14) + \frac{\sqrt[3]{8-x}}{x+8} \quad ??? \quad (-\infty, -8) \cup (-8, -7) \cup (2, \infty)$$

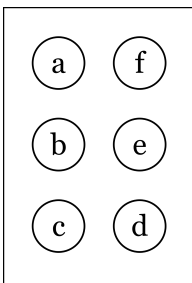
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_7(2-\frac{2}{3}t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -4, \frac{3}{2} \right\rangle \cup \left(\frac{3}{2}, 3 \right)$$

$$(d) \quad u(x) = \frac{-8x-1}{-2x-5} \quad \dots \quad ??? \quad \dots \quad (-\infty, -\frac{5}{2}) \cup (-\frac{1}{8}, \infty)$$

$$(e) \quad \varphi(t) = 2t^3 - 4t^2 + \frac{3}{t} + 3, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{176}{27}$$

$$(f) \quad w(x) = \frac{x}{9} + \frac{9}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



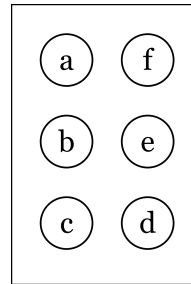
Braille letter

Function of one variable, group *Kappa* κ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -3 \arcsin\left(\frac{5-8x}{4}\right) - \frac{e^{-4x+2}}{\sqrt{32x^2-20x}}$... ??? ... $\left(\frac{5}{8}, \frac{9}{8}\right)$
 (b) $g(x) = \frac{1}{3} \ln(x^2 - 7x + 6) + \frac{\sqrt[3]{2-x}}{x-7}$??? $(-\infty, 1) \cup (6, 7) \cup (7, \infty)$
 (c) $h(t) = \frac{\sqrt{25-t^2}}{\log_3(6-\frac{3}{2}t)}$??? $\left\langle -5, \frac{10}{3} \right\rangle \cup \left(\frac{10}{3}, 4\right)$
 (d) $u(x) = \frac{-x-1}{-x-8}$??? $(-\infty, -8) \cup (-1, \infty)$
 (e) $\varphi(t) = -3t^3 + 4t^2 + \frac{7}{t} - 7$, $t_0 = -\frac{1}{2}$??? $-\frac{73}{8}$
 (f) $w(x) = \sqrt[3]{-8x^2-4x-2}$??? 0

1.



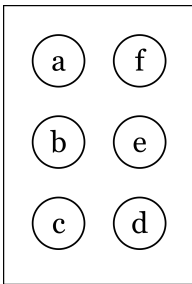
Braille letter

Function of one variable, group *Kappa* κ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 4 \arcsin\left(\frac{7-4x}{3}\right) - \frac{e^{4x-3}}{\sqrt{16x^2-28x}}$??? $\left(\frac{7}{4}, \frac{5}{2}\right)$
 (b) $g(x) = \frac{1}{5} \ln(x^2 + 4x - 21) + \frac{\sqrt[3]{-3-x}}{x+8}$??? $(-\infty, -8) \cup (-8, -7) \cup (3, \infty)$
 (c) $h(t) = \frac{\sqrt{64-t^2}}{\log_2(2-\frac{2}{7}t)}$??? $\left\langle -8, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 7\right)$
 (d) $u(x) = \frac{5x+6}{-6x+4}$??? $\left\langle \frac{-6}{5}, \frac{2}{3} \right\rangle$
 (e) $\varphi(t) = 5t^3 - 2t^2 + \frac{4}{t} - 3$, $t_0 = -\frac{1}{2}$??? $-\frac{49}{8}$
 (f) $w(x) = \frac{x}{5} + \frac{5}{x}$??? 2

2.



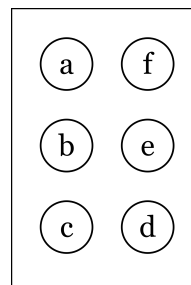
Braille letter

Function of one variable, group *Kappa* κ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -3 \arcsin\left(\frac{4-4x}{3}\right) - \frac{e^{-3x+8}}{\sqrt{-24x^2+24x}}$. ??? . $\left(1, \frac{7}{4}\right)$
 (b) $g(x) = \frac{1}{4} \ln(x^2 - 5x + 4) + \frac{\sqrt[3]{9-x}}{x-5}$??? $(-\infty, 1) \cup (4, 5) \cup (5, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_8(3+\frac{3}{5}t)}$??? $\left(-5, -\frac{10}{3}\right) \cup \left(-\frac{10}{3}, 6\right)$
 (d) $u(x) = \frac{-x-2}{-x+5}$??? $(-\infty, -2) \cup (5, \infty)$
 (e) $\varphi(t) = -2t^3 - 4t^2 + \frac{5}{t} - 8$, $t_0 = -\frac{1}{2}$??? $-\frac{45}{4}$
 (f) $w(x) = -2x^3 - 14x^2 - 12x$??? 3

3.



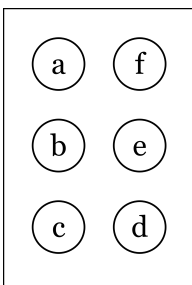
Braille letter

Function of one variable, group *Kappa* κ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 6 \arccos\left(\frac{7-3x}{3}\right) - \frac{e^{-2x-8}}{\sqrt{9x^2-21x}}$... ??? ... $\left(\frac{7}{3}, \frac{10}{3}\right)$
 (b) $g(x) = \frac{1}{3} \ln(x^2 - 9x + 8) + \frac{\sqrt[3]{3-x}}{x-9}$??? $(-\infty, 1) \cup (8, 9) \cup (9, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_6(5-\frac{5}{2}t)}$??? $\left\langle -3, \frac{8}{5} \right\rangle \cup \left(\frac{8}{5}, 2\right)$
 (d) $u(x) = \frac{6x-2}{5x+9}$??? $(-\infty, \frac{-9}{5}) \cup \left(\frac{1}{3}, \infty\right)$
 (e) $\varphi(t) = t^3 - 3t^2 + \frac{3}{t} + 1$, $t_0 = -\frac{1}{2}$??? $-\frac{47}{8}$
 (f) $w(x) = -4x^3 + 15x^2 + 4x$??? 1

4.



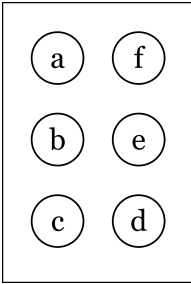
Braille letter

Function of one variable, group *Lambda* λ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -4 \arccos\left(\frac{5-7x}{2}\right) - \frac{e^{2x+1}}{\sqrt{35x^2 - 25x}} \dots ??? \dots \left(\frac{5}{7}, 1\right)$
 (b) $g(x) = \frac{1}{5} \ln(x^2 + x - 12) + \frac{\sqrt[3]{-1-x}}{x-4} \dots (-\infty, -4) \cup (3, 4) \cup (4, \infty)$
 (c) $h(t) = \frac{\sqrt{16-t^2}}{\log_6(6-2t)} \dots \dots \dots \langle -4, \frac{5}{2} \rangle \cup \left(\frac{5}{2}, 3\right)$
 (d) $u(x) = \frac{-4x-2}{-2x-4} \dots \dots \dots (-\infty, -2) \cup \left(\frac{-1}{2}, \infty\right)$
 (e) $\varphi(t) = t^3 - 3t^2 + \frac{8}{t} - 7$, $t_0 = -\frac{1}{3} \dots \dots \dots -\frac{271}{27}$
 (f) $w(x) = \sqrt[3]{-7x^2 + 5x - 4} \dots \dots \dots 1$

1.



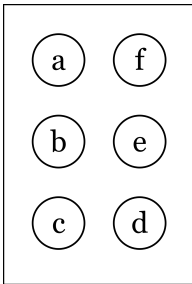
Braille letter

Function of one variable, group *Lambda* λ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 5 \arccos\left(\frac{9-8x}{6}\right) - \frac{e^{x-2}}{\sqrt{-32x^2 + 36x}} \dots ??? \dots \left\langle \frac{3}{8}, \frac{9}{8} \right\rangle$
 (b) $g(x) = \frac{1}{5} \ln(x^2 - 4x - 12) + \frac{\sqrt[3]{1-x}}{x-7} \dots \dots \dots (-\infty, -2) \cup (6, 7) \cup (7, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_2(7+\frac{7}{2}t)} \dots \dots \dots \left(-2, -\frac{12}{7}\right) \cup \left(-\frac{12}{7}, 3\right)$
 (d) $u(x) = \frac{-5x+1}{7x+3} \dots \dots \dots \left(\frac{-3}{7}, \frac{1}{5}\right)$
 (e) $\varphi(t) = -t^3 - 5t^2 + \frac{4}{t} - 4$, $t_0 = -\frac{1}{4} \dots \dots \dots -\frac{1299}{64}$
 (f) $w(x) = \sqrt[3]{5x^2 - 2x + 3} \dots \dots \dots 1$

2.



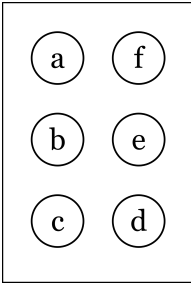
Braille letter

Function of one variable, group *Lambda* λ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -4 \arcsin\left(\frac{8-2x}{5}\right) - \frac{e^{7x-8}}{\sqrt{-12x^2 + 48x}} \dots \dots \dots \left\langle \frac{3}{2}, 4 \right\rangle$
 (b) $g(x) = \frac{1}{5} \ln(x^2 + 13x + 40) + \frac{\sqrt[3]{7-x}}{x+9} \dots \dots \dots (-\infty, -9) \cup (-9, -8) \cup (-5, \infty)$
 (c) $h(t) = \frac{\sqrt{25-t^2}}{\log_4(8-2t)} \dots \dots \dots \left\langle -5, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 4\right)$
 (d) $u(x) = \frac{5x-2}{3x-1} \dots \dots \dots (-\infty, \frac{1}{3}) \cup \left(\frac{2}{5}, \infty\right)$
 (e) $\varphi(t) = 7t^3 - 6t^2 + \frac{3}{t} - 5$, $t_0 = -\frac{1}{3} \dots \dots \dots -\frac{187}{27}$
 (f) $w(x) = 6x^3 + 10x^2 - 4x \dots \dots \dots 2$

3.



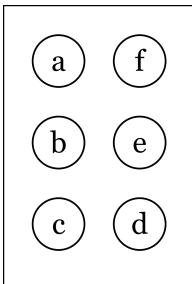
Braille letter

Function of one variable, group *Lambda* λ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 7 \arcsin\left(\frac{4-8x}{3}\right) - \frac{e^{-7x-3}}{\sqrt{24x^2 - 12x}} \dots \dots \dots \left(\frac{1}{2}, \frac{7}{8}\right)$
 (b) $g(x) = \frac{1}{7} \ln(x^2 - 13x + 42) + \frac{\sqrt[3]{-5-x}}{x-5} \dots \dots \dots (-\infty, 5) \cup (5, 6) \cup (7, \infty)$
 (c) $h(t) = \frac{\sqrt{49-t^2}}{\log_2(2-\frac{1}{3}t)} \dots \dots \dots \langle -7, 3 \rangle \cup (3, 6)$
 (d) $u(x) = \frac{-x-2}{2x-6} \dots \dots \dots \langle -2, 3 \rangle$
 (e) $\varphi(t) = -t^3 + 3t^2 + \frac{7}{t} + 6$, $t_0 = -\frac{1}{2} \dots \dots \dots \frac{27}{8}$
 (f) $w(x) = \frac{x}{4} + \frac{4}{x} \dots \dots \dots 0$

4.



Braille letter

Function of one variable, group $Mu \mu$ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -4 \arccos\left(\frac{7-x}{2}\right) - \frac{e^{-3x+8}}{\sqrt{7x^2-49x}}$??? $(7, 9)$

(b) $g(x) = \frac{1}{4} \ln(x^2 + 9x + 18) + \frac{\sqrt[3]{1-x}}{x+7}$??? $(-\infty, -7) \cup (-7, -6) \cup (-3, \infty)$

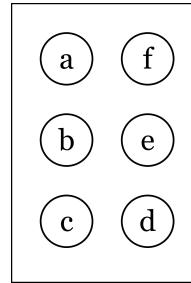
(c) $h(t) = \frac{\sqrt{25-t^2}}{\log_2(1+\frac{1}{4}t)}$??? $(-4, 0) \cup (0, 5)$

(d) $u(x) = \frac{-6x-2}{-x-4}$??? $(-\infty, -4) \cup (\frac{-1}{3}, \infty)$

(e) $\varphi(t) = -t^3 - t^2 + \frac{6}{t} + 5$, $t_0 = -\frac{1}{3}$??? $\frac{79}{27}$

(f) $w(x) = 6x^3 - 39x^2 + 18x$??? 3

1.



Braille letter

Function of one variable, group $Mu \mu$ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 5 \arccos\left(\frac{6-5x}{4}\right) - \frac{e^{-x-1}}{\sqrt{5x^2-6x}}$??? $(\frac{6}{5}, 2)$

(b) $g(x) = \frac{1}{7} \ln(x^2 + 2x - 24) + \frac{\sqrt[3]{-3-x}}{x-5}$??? $(-\infty, -6) \cup (4, 5) \cup (5, \infty)$

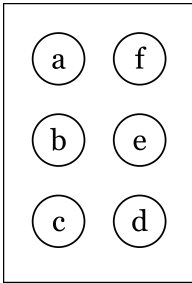
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_8(3+\frac{3}{5}t)}$??? $(-5, -\frac{10}{3}) \cup (-\frac{10}{3}, 6)$

(d) $u(x) = \frac{6x-2}{-9x+9}$??? $(\frac{1}{3}, 1)$

(e) $\varphi(t) = 2t^3 - 2t^2 - \frac{1}{t} - 5$, $t_0 = -\frac{1}{2}$??? $-\frac{21}{4}$

(f) $w(x) = 6x^3 - 42x^2 + 36x$??? 1

2.



Braille letter

Function of one variable, group $Mu \mu$ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -9 \arcsin\left(\frac{6-3x}{4}\right) - \frac{e^{6x+4}}{\sqrt{-9x^2+18x}}$??? $(2, \frac{10}{3})$

(b) $g(x) = \frac{1}{9} \ln(x^2 + 13x + 42) + \frac{\sqrt[3]{2-x}}{x+8}$??? $(-\infty, -8) \cup (-8, -7) \cup (-6, \infty)$

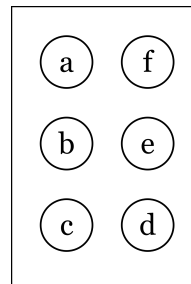
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_6(2+\frac{2}{5}t)}$??? $(-5, -\frac{5}{2}) \cup (-\frac{5}{2}, 6)$

(d) $u(x) = \frac{-5x-3}{-8x+2}$??? $(-\infty, \frac{-3}{5}) \cup (\frac{1}{4}, \infty)$

(e) $\varphi(t) = -5t^3 + 7t^2 + \frac{5}{t} - 6$, $t_0 = -\frac{1}{2}$??? $-\frac{109}{8}$

(f) $w(x) = -3x^3 + 3x^2 + 6x$??? 3

3.



Braille letter

Function of one variable, group $Mu \mu$ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -7 \arcsin\left(\frac{6-5x}{2}\right) - \frac{e^{-x-5}}{\sqrt{-10x^2+12x}}$??? $(\frac{4}{5}, \frac{6}{5})$

(b) $g(x) = \frac{1}{3} \ln(x^2 + 8x + 15) + \frac{\sqrt[3]{1-x}}{x+6}$??? $(-\infty, -6) \cup (-6, -5) \cup (-3, \infty)$

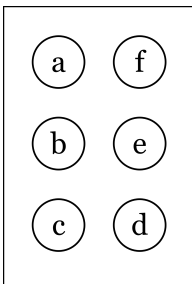
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_8(1-\frac{1}{5}t)}$??? $(-6, 0) \cup (0, 5)$

(d) $u(x) = \frac{4x+7}{-2x-8}$??? $(-4, \frac{-7}{4})$

(e) $\varphi(t) = 8t^3 - 7t^2 - \frac{1}{t} - 7$, $t_0 = -\frac{1}{3}$??? $-\frac{137}{27}$

(f) $w(x) = -8x^3 - 6x^2 + 2x$??? 2

4.



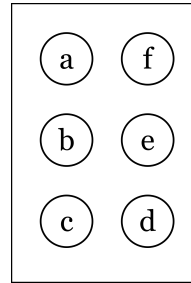
Braille letter

Function of one variable, group $Nu \nu$ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -5 \arcsin\left(\frac{8-7x}{6}\right) - \frac{e^{2x+5}}{\sqrt{49x^2 - 56x}} \dots ??? \dots \left(\frac{8}{7}, 2\right)$
 (b) $g(x) = \frac{1}{7} \ln(x^2 + 2x - 8) + \frac{\sqrt[3]{-2-x}}{x-3} ??? (-\infty, -4) \cup (2, 3) \cup (3, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_4(4+\frac{4}{5}t)} \dots ??? \dots (-5, -\frac{15}{4}) \cup (-\frac{15}{4}, 6)$
 (d) $u(x) = \frac{-2x+1}{-5x+1} \dots ??? \dots (-\infty, \frac{1}{5}) \cup (\frac{1}{2}, \infty)$
 (e) $\varphi(t) = 7t^3 + 5t^2 - \frac{7}{t} - 2, t_0 = -\frac{1}{2} \dots ??? \dots \frac{15}{8}$
 (f) $w(x) = 4x^3 + 35x^2 + 24x \dots ??? \dots 1$

1.



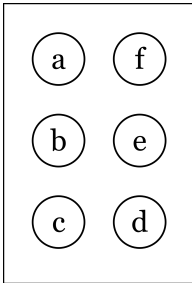
Braille letter

Function of one variable, group $Nu \nu$ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -6 \arccos\left(\frac{8-3x}{2}\right) - \frac{e^{6x-2}}{\sqrt{3x^2 - 8x}} \dots ??? \dots \left\langle 2, \frac{8}{3} \right\rangle$
 (b) $g(x) = \frac{1}{5} \ln(x^2 + 4x + 3) + \frac{\sqrt[3]{-5-x}}{x+4} ??? (-\infty, -4) \cup (-4, -3) \cup (-1, \infty)$
 (c) $h(t) = \frac{\sqrt{25-t^2}}{\log_2(1-\frac{1}{4}t)} \dots ??? \dots \langle -5, 0 \rangle \cup (0, 4)$
 (d) $u(x) = \frac{5x-1}{6x+1} \dots ??? \dots (-\infty, \frac{-1}{6}) \cup \langle \frac{1}{5}, \infty \rangle$
 (e) $\varphi(t) = 3t^3 + 4t^2 - \frac{1}{t} - 4, t_0 = -\frac{1}{3} \dots ??? \dots -\frac{2}{3}$
 (f) $w(x) = \sqrt[7]{-3x^2 + 7x - 6} \dots ??? \dots 0$

2.



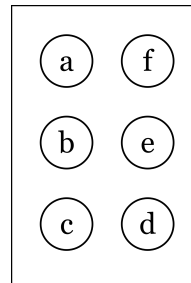
Braille letter

Function of one variable, group $Nu \nu$ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -5 \arccos\left(\frac{3-2x}{2}\right) - \frac{e^{4x-4}}{\sqrt{-6x^2 + 9x}} \dots ??? \dots \left\langle \frac{1}{2}, \frac{3}{2} \right\rangle$
 (b) $g(x) = \frac{1}{3} \ln(x^2 - 11x + 30) + \frac{\sqrt[3]{-5-x}}{x-7} ??? (-\infty, 5) \cup (6, 7) \cup (7, \infty)$
 (c) $h(t) = \frac{\sqrt{4-t^2}}{\log_4(8-8t)} \dots ??? \dots \langle -2, \frac{7}{8} \rangle \cup (\frac{7}{8}, 1)$
 (d) $u(x) = \frac{-8x+5}{7x+4} \dots ??? \dots (-\frac{4}{7}, \frac{5}{8})$
 (e) $\varphi(t) = 6t^3 - t^2 - \frac{2}{t} + 7, t_0 = -\frac{1}{3} \dots ??? \dots \frac{38}{3}$
 (f) $w(x) = \frac{x}{2} + \frac{2}{x} \dots ??? \dots 1$

3.



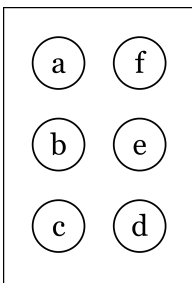
Braille letter

Function of one variable, group $Nu \nu$ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 3 \arccos\left(\frac{4-5x}{2}\right) - \frac{e^{-9x-1}}{\sqrt{15x^2 - 12x}} \dots ??? \dots \left(\frac{4}{5}, \frac{6}{5}\right)$
 (b) $g(x) = \frac{1}{4} \ln(x^2 + x - 2) + \frac{\sqrt[3]{1-x}}{x-2} ??? (-\infty, -2) \cup (1, 2) \cup (2, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_3(9-\frac{9}{2}t)} \dots ??? \dots \langle -3, \frac{16}{9} \rangle \cup (\frac{16}{9}, 2)$
 (d) $u(x) = \frac{2x-1}{6x+5} \dots ??? \dots (-\infty, \frac{-5}{6}) \cup \langle \frac{1}{2}, \infty \rangle$
 (e) $\varphi(t) = -6t^3 - 8t^2 + \frac{3}{t} - 4, t_0 = -\frac{1}{4} \dots ??? \dots -\frac{165}{32}$
 (f) $w(x) = -\frac{x}{2} - \frac{2}{x} \dots ??? \dots 0$

4.



Braille letter

Function of one variable, group $Xi \xi$ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = \arcsin\left(\frac{3-7x}{2}\right) - \frac{e^{-5x-5}}{\sqrt{7x^2-3x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{3}{7}, \frac{5}{7}\right)$$

$$(b) \quad g(x) = \frac{1}{6} \ln(x^2 - 8x + 15) + \frac{\sqrt[3]{4-x}}{x-6} \quad ??? \quad (-\infty, 3) \cup (5, 6) \cup (6, \infty)$$

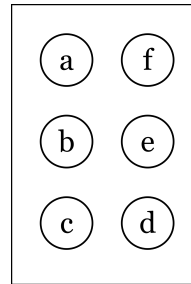
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_6(6+6t)} \quad \dots \quad ??? \quad \dots \quad \left(-1, -\frac{5}{6}\right) \cup \left(-\frac{5}{6}, 2\right)$$

$$(d) \quad u(x) = \frac{-x-8}{x-3} \quad \dots \quad ??? \quad \dots \quad \langle -8, 3 \rangle$$

$$(e) \quad \varphi(t) = 4t^3 + 7t^2 + \frac{3}{t} - 1, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{101}{8}$$

$$(f) \quad w(x) = \sqrt[5]{5x^2 - 6x + 2} \quad \dots \quad ??? \quad \dots \quad 0$$

1.



Braille letter

Function of one variable, group $Xi \xi$ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 2 \arccos\left(\frac{7-x}{2}\right) - \frac{e^{3x+1}}{\sqrt{-6x^2+42x}} \quad \dots \quad ??? \quad \dots \quad \langle 5, 7 \rangle$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 + x - 42) + \frac{\sqrt[3]{3-x}}{x+8} \quad ??? \quad (-\infty, -8) \cup (-8, -7) \cup (6, \infty)$$

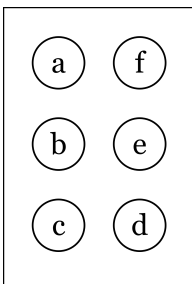
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_4(5-\frac{5}{2}t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -3, \frac{8}{5} \right\rangle \cup \left(\frac{8}{5}, 2 \right)$$

$$(d) \quad u(x) = \frac{-2x+5}{-x-4} \quad \dots \quad ??? \quad \dots \quad (-\infty, -4) \cup \left\langle \frac{5}{2}, \infty \right\rangle$$

$$(e) \quad \varphi(t) = -2t^3 + 3t^2 + \frac{3}{t} - 6, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{394}{27}$$

$$(f) \quad w(x) = \frac{x}{2} + \frac{2}{x} \quad \dots \quad ??? \quad \dots \quad 1$$

2.



Braille letter

Function of one variable, group $Xi \xi$ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arcsin\left(\frac{9-x}{8}\right) - \frac{e^{4x+2}}{\sqrt{4x^2-36x}} \quad \dots \quad ??? \quad \dots \quad (9, 17)$$

$$(b) \quad g(x) = \frac{1}{7} \ln(x^2 + 3x - 4) + \frac{\sqrt[3]{1-x}}{x-2} \quad ??? \quad (-\infty, -4) \cup (1, 2) \cup (2, \infty)$$

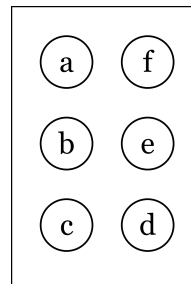
$$(c) \quad h(t) = \frac{\sqrt{49-t^2}}{\log_3(1+\frac{1}{6}t)} \quad \dots \quad ??? \quad \dots \quad (-6, 0) \cup (0, 7)$$

$$(d) \quad u(x) = \frac{3x+6}{x-5} \quad \dots \quad ??? \quad \dots \quad (-\infty, -2) \cup (5, \infty)$$

$$(e) \quad \varphi(t) = 4t^3 - 4t^2 - \frac{2}{t} - 4, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{9}{2}$$

$$(f) \quad w(x) = \sqrt[3]{-2x^2 + 2x - 1} \quad \dots \quad ??? \quad \dots \quad 2$$

3.



Braille letter

Function of one variable, group $Xi \xi$ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arcsin\left(\frac{9-2x}{5}\right) - \frac{e^{-5x-6}}{\sqrt{12x^2-54x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{9}{2}, 7\right)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 - 2x - 24) + \frac{\sqrt[3]{2-x}}{x+5} \quad ??? \quad (-\infty, -5) \cup (-5, -4) \cup (6, \infty)$$

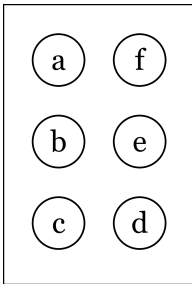
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_2(1-t)} \quad \dots \quad ??? \quad \dots \quad \langle -2, 0 \rangle \cup (0, 1)$$

$$(d) \quad u(x) = \frac{-2x+7}{2x-1} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{1}{2}, \frac{7}{2} \right\rangle$$

$$(e) \quad \varphi(t) = -3t^3 - 2t^2 + \frac{3}{t} - 3, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{109}{9}$$

$$(f) \quad w(x) = \sqrt[3]{-4x^2 - 2x - 1} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



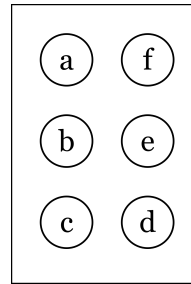
Braille letter

Function of one variable, group *Omicron o* -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -4 \arccos\left(\frac{5-5x}{2}\right) - \frac{e^{-8x-5}}{\sqrt{20x^2-20x}}$??? $\left\langle \frac{3}{5}, 1 \right\rangle$
 (b) $g(x) = \frac{1}{5} \ln(x^2 + 5x + 4) + \frac{\sqrt[3]{2-x}}{x+5}$??? $(-\infty, -5) \cup (-5, -4) \cup (-1, \infty)$
 (c) $h(t) = \frac{\sqrt{25-t^2}}{\log_4(2-\frac{1}{2}t)}$??? $\langle -5, 2 \rangle \cup (2, 4)$
 (d) $u(x) = \frac{5x-4}{5x+1}$??? $(-\infty, \frac{-1}{5}) \cup \langle \frac{4}{5}, \infty \rangle$
 (e) $\varphi(t) = 5t^3 - 6t^2 + \frac{1}{t} - 3$, $t_0 = -\frac{1}{3}$??? $-\frac{185}{27}$
 (f) $w(x) = -6x^3 + 29x^2 - 20x$??? 3

1.



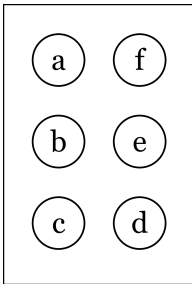
Braille letter

Function of one variable, group *Omicron o* -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = -7 \arcsin\left(\frac{4-7x}{2}\right) - \frac{e^{-7x-9}}{\sqrt{21x^2-12x}}$.. ??? .. $\left\langle \frac{2}{7}, \frac{4}{7} \right\rangle$
 (b) $g(x) = \frac{1}{6} \ln(x^2 + x - 2) + \frac{\sqrt[3]{3-x}}{x-2}$??? $(-\infty, -2) \cup (1, 2) \cup (2, \infty)$
 (c) $h(t) = \frac{\sqrt{49-t^2}}{\log_8(2+\frac{1}{3}t)}$??? $(-6, -3) \cup (-3, 7)$
 (d) $u(x) = \frac{-5x+1}{-x-2}$??? $(-\infty, -2) \cup \langle \frac{1}{5}, \infty \rangle$
 (e) $\varphi(t) = 7t^3 - 3t^2 - \frac{1}{t} + 6$, $t_0 = -\frac{1}{3}$??? $\frac{155}{27}$
 (f) $w(x) = \sqrt[5]{-7x^2+5x-4}$??? 0

2.



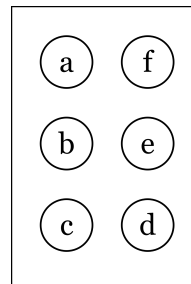
Braille letter

Function of one variable, group *Omicron o* -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 6 \arccos\left(\frac{7-5x}{4}\right) - \frac{e^{-2x+3}}{\sqrt{20x^2-28x}}$??? $\left\langle \frac{7}{5}, \frac{11}{5} \right\rangle$
 (b) $g(x) = \frac{1}{4} \ln(x^2 + 3x - 28) + \frac{\sqrt[3]{-5-x}}{x-5}$??? $(-\infty, -7) \cup (4, 5) \cup (5, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_5(3-\frac{3}{5}t)}$??? $\langle -6, \frac{10}{3} \rangle \cup (\frac{10}{3}, 5)$
 (d) $u(x) = \frac{-8x-3}{3x+8}$??? $(\frac{-8}{3}, \frac{-3}{8})$
 (e) $\varphi(t) = -7t^3 + 2t^2 + \frac{6}{t} - 3$, $t_0 = -\frac{1}{4}$??? $-\frac{273}{64}$
 (f) $w(x) = -x^3 - x^2 + 12x$??? 1

3.



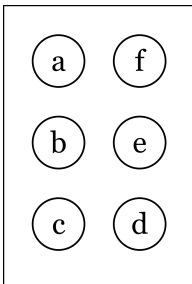
Braille letter

Function of one variable, group *Omicron o* -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = \arcsin\left(\frac{6-6x}{5}\right) - \frac{e^{7x-1}}{\sqrt{-24x^2+24x}}$??? $\langle \frac{1}{6}, 1 \rangle$
 (b) $g(x) = \frac{1}{3} \ln(x^2 - 9x + 20) + \frac{\sqrt[3]{-2-x}}{x-6}$??? $(-\infty, 4) \cup (5, 6) \cup (6, \infty)$
 (c) $h(t) = \frac{\sqrt{9-t^2}}{\log_2(4+2t)}$??? $(-2, -\frac{3}{2}) \cup (-\frac{3}{2}, 3)$
 (d) $u(x) = \frac{-7x+1}{x-7}$??? $(\frac{1}{7}, 7)$
 (e) $\varphi(t) = -6t^3 + 3t^2 - \frac{3}{t} + 9$, $t_0 = -\frac{1}{4}$??? $\frac{681}{32}$
 (f) $w(x) = \frac{x}{2} + \frac{2}{x}$??? 0

4.



Braille letter

Function of one variable, group $Pi \pi$ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arccos\left(\frac{4-2x}{3}\right) - \frac{e^{-4x-5}}{\sqrt{-10x^2+20x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{1}{2}, 2 \right\rangle$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + 4x - 45) + \frac{\sqrt[3]{-4-x}}{x-6} \quad ??? \quad (-\infty, -9) \cup (5, 6) \cup (6, \infty)$$

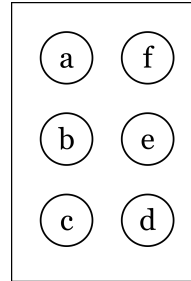
$$(c) \quad h(t) = \frac{\sqrt{49-t^2}}{\log_2(1-\frac{1}{6}t)} \quad \dots \quad ??? \quad \dots \quad \langle -7, 0 \rangle \cup (0, 6)$$

$$(d) \quad u(x) = \frac{9x+3}{-5x+4} \quad \dots \quad ??? \quad \dots \quad \left\langle -\frac{1}{3}, \frac{4}{5} \right\rangle$$

$$(e) \quad \varphi(t) = 2t^3 + 4t^2 - \frac{1}{t} - 4, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{89}{27}$$

$$(f) \quad w(x) = -\frac{x}{4} - \frac{4}{x} \quad \dots \quad ??? \quad \dots \quad 2$$

1.



Braille letter

Function of one variable, group $Pi \pi$ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -7 \arcsin\left(\frac{6-x}{5}\right) - \frac{e^{-x-2}}{\sqrt{5x^2-30x}} \quad \dots \quad ??? \quad \dots \quad \langle 1, 6 \rangle$$

$$(b) \quad g(x) = \frac{1}{9} \ln(x^2 - 6x - 27) + \frac{\sqrt[3]{1-x}}{x+4} \quad ??? \quad (-\infty, -4) \cup (-4, -3) \cup (9, \infty)$$

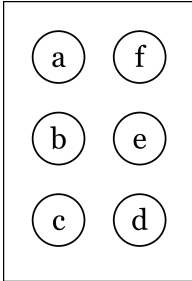
$$(c) \quad h(t) = \frac{\sqrt{81-t^2}}{\log_7(2-\frac{1}{4}t)} \quad \dots \quad ??? \quad \dots \quad \langle -9, 4 \rangle \cup (4, 8)$$

$$(d) \quad u(x) = \frac{-6x+1}{-x+3} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{1}{6}) \cup (3, \infty)$$

$$(e) \quad \varphi(t) = 7t^3 - 8t^2 - \frac{2}{t} - 1, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{40}{27}$$

$$(f) \quad w(x) = \sqrt[3]{2x^2-x+4} \quad \dots \quad ??? \quad \dots \quad 0$$

2.



Braille letter

Function of one variable, group $Pi \pi$ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 3 \arccos\left(\frac{8-x}{2}\right) - \frac{e^{-2x-8}}{\sqrt{-3x^2+24x}} \quad \dots \quad ??? \quad \dots \quad \langle 6, 8 \rangle$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 + 4x - 21) + \frac{\sqrt[3]{-7-x}}{x-4} \quad ??? \quad (-\infty, -7) \cup (3, 4) \cup (4, \infty)$$

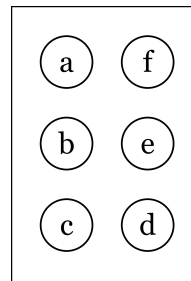
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_2(5+5t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{4}{5}) \cup (-\frac{4}{5}, 2)$$

$$(d) \quad u(x) = \frac{6x+7}{-x+5} \quad \dots \quad ??? \quad \dots \quad \left\langle -\frac{7}{6}, 5 \right\rangle$$

$$(e) \quad \varphi(t) = -4t^3 + 7t^2 + \frac{6}{t} + 3, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{380}{27}$$

$$(f) \quad w(x) = \sqrt[3]{-7x^2+3x-3} \quad \dots \quad ??? \quad \dots \quad 2$$

3.



Braille letter

Function of one variable, group $Pi \pi$ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = \arcsin\left(\frac{9-7x}{2}\right) - \frac{e^{6x-5}}{\sqrt{-49x^2+63x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{9}{7}, \frac{11}{7}\right)$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 - 3x - 10) + \frac{\sqrt[3]{4-x}}{x-6} \quad ??? \quad (-\infty, -2) \cup (5, 6) \cup (6, \infty)$$

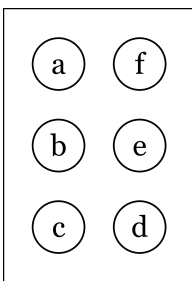
$$(c) \quad h(t) = \frac{\sqrt{25-t^2}}{\log_7(2+\frac{1}{2}t)} \quad \dots \quad ??? \quad \dots \quad (-4, -2) \cup (-2, 5)$$

$$(d) \quad u(x) = \frac{3x+3}{4x-8} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup (2, \infty)$$

$$(e) \quad \varphi(t) = -8t^3 - t^2 - \frac{1}{t} - 5, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{15}{16}$$

$$(f) \quad w(x) = \sqrt[3]{-3x^2-5x-6} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



Braille letter

Function of one variable, group *Rho* ρ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -7 \arcsin\left(\frac{7-x}{5}\right) - \frac{e^{-4x+6}}{\sqrt{-6x^2+42x}} \quad \dots \quad ??? \quad \dots \quad \langle 2, 7 \rangle$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 5x + 4) + \frac{\sqrt[3]{-6-x}}{x-5} \quad ??? \quad (-\infty, 1) \cup (4, 5) \cup (5, \infty)$$

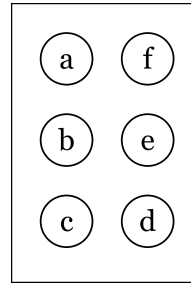
$$(c) \quad h(t) = \frac{\sqrt{64-t^2}}{\log_2(8+\frac{8}{7}t)} \quad \dots \quad ??? \quad \dots \quad (-7, -\frac{49}{8}) \cup (-\frac{49}{8}, 8)$$

$$(d) \quad u(x) = \frac{x-2}{9x-3} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{1}{3}) \cup (2, \infty)$$

$$(e) \quad \varphi(t) = -6t^3 + t^2 + \frac{1}{t} - 3, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{99}{32}$$

$$(f) \quad w(x) = 7x^3 + 44x^2 + 12x \quad \dots \quad ??? \quad \dots \quad 1$$

1.



Braille letter

Function of one variable, group *Rho* ρ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arccos\left(\frac{6-2x}{2}\right) - \frac{e^{-2x-8}}{\sqrt{-8x^2+24x}} \quad \dots \quad ??? \quad \dots \quad \langle 2, 3 \rangle$$

$$(b) \quad g(x) = \frac{1}{6} \ln(x^2 - 5x + 6) + \frac{\sqrt[3]{3-x}}{x-1} \quad ??? \quad (-\infty, 1) \cup (1, 2) \cup (3, \infty)$$

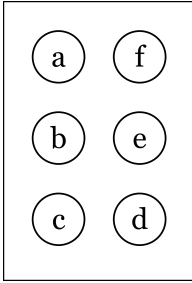
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_3(8+4t)} \quad \dots \quad ??? \quad \dots \quad (-2, -\frac{7}{4}) \cup (-\frac{7}{4}, 3)$$

$$(d) \quad u(x) = \frac{7x-5}{5x-4} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{5}{7}) \cup (\frac{4}{5}, \infty)$$

$$(e) \quad \varphi(t) = 8t^3 - 2t^2 - \frac{3}{t} + 5, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad 5$$

$$(f) \quad w(x) = -3x^3 - 16x^2 + 35x \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Rho* ρ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arccos\left(\frac{5-5x}{2}\right) - \frac{e^{-2x+7}}{\sqrt{5x^2-5x}} \quad \dots \quad ??? \quad \dots \quad (1, \frac{7}{5})$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + 5x + 6) + \frac{\sqrt[3]{-2-x}}{x+1} \quad ??? \quad (-\infty, -3) \cup (-2, -1) \cup (-1, \infty)$$

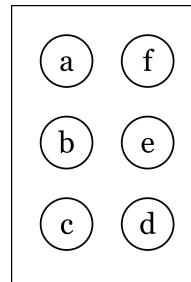
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_3(3+\frac{3}{2}t)} \quad \dots \quad ??? \quad \dots \quad (-2, -\frac{4}{3}) \cup (-\frac{4}{3}, 3)$$

$$(d) \quad u(x) = \frac{-6x-6}{2x-2} \quad \dots \quad ??? \quad \dots \quad \langle -1, 1 \rangle$$

$$(e) \quad \varphi(t) = -3t^3 - 8t^2 - \frac{5}{t} - 1, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad -\frac{1}{8}$$

$$(f) \quad w(x) = \frac{x}{8} + \frac{8}{x} \quad \dots \quad ??? \quad \dots \quad 2$$

3.



Braille letter

Function of one variable, group *Rho* ρ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 4 \arccos\left(\frac{5-4x}{2}\right) - \frac{e^{-5x+1}}{\sqrt{-20x^2+25x}} \quad \dots \quad ??? \quad \dots \quad \langle \frac{3}{4}, \frac{5}{4} \rangle$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 - 4x + 3) + \frac{\sqrt[3]{-2-x}}{x-4} \quad ??? \quad (-\infty, 1) \cup (3, 4) \cup (4, \infty)$$

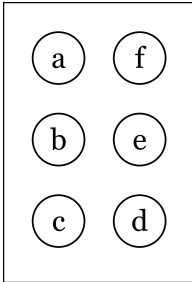
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_2(7+7t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{6}{7}) \cup (-\frac{6}{7}, 2)$$

$$(d) \quad u(x) = \frac{-3x+1}{-4x+2} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{1}{3}) \cup (\frac{1}{2}, \infty)$$

$$(e) \quad \varphi(t) = -7t^3 - 8t^2 - \frac{6}{t} + 5, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad \frac{127}{8}$$

$$(f) \quad w(x) = \sqrt[7]{8x^2+5x+3} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



Braille letter

Function of one variable, group *Sigma* σ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 5 \arccos\left(\frac{7-x}{2}\right) - \frac{e^{3x-4}}{\sqrt{-x^2+7x}}$??? $\langle 5, 7 \rangle$

(b) $g(x) = \frac{1}{5} \ln(x^2 + 5x - 6) + \frac{\sqrt[3]{-9-x}}{x-2}$??? $(-\infty, -6) \cup (1, 2) \cup (2, \infty)$

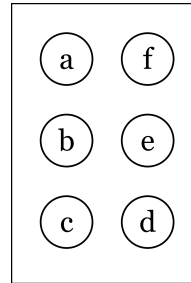
(c) $h(t) = \frac{\sqrt{25-t^2}}{\log_3(5+\frac{5}{4}t)}$??? $(-4, -\frac{16}{5}) \cup (-\frac{16}{5}, 5)$

(d) $u(x) = \frac{-3x+1}{7x+6}$??? $(-\frac{6}{7}, \frac{1}{3})$

(e) $\varphi(t) = 4t^3 + 7t^2 - \frac{1}{t} - 6$, $t_0 = -\frac{1}{2}$??? $-\frac{11}{4}$

(f) $w(x) = \sqrt[3]{-5x^2-3x-4}$??? 0

1.



Braille letter

Function of one variable, group *Sigma* σ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -5 \arccos\left(\frac{8-3x}{4}\right) - \frac{e^{-3x-3}}{\sqrt{-6x^2+16x}}$??? $\langle \frac{4}{3}, \frac{8}{3} \rangle$

(b) $g(x) = \frac{1}{2} \ln(x^2 + 15x + 56) + \frac{\sqrt[3]{3-x}}{x+6}$??? $(-\infty, -8) \cup (-7, -6) \cup (-6, \infty)$

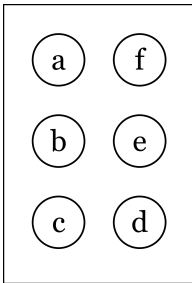
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_7(8+\frac{8}{5}t)}$??? $(-5, -\frac{35}{8}) \cup (-\frac{35}{8}, 6)$

(d) $u(x) = \frac{x-2}{-x-4}$??? $(-4, 2)$

(e) $\varphi(t) = -2t^3 + t^2 + \frac{6}{t} - 2$, $t_0 = -\frac{1}{3}$??? $-\frac{535}{27}$

(f) $w(x) = \frac{x}{4} + \frac{4}{x}$??? 1

2.



Braille letter

Function of one variable, group *Sigma* σ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 9 \arccos\left(\frac{4-2x}{3}\right) - \frac{e^{7x+2}}{\sqrt{2x^2-4x}}$??? $(2, \frac{7}{2})$

(b) $g(x) = \frac{1}{2} \ln(x^2 - 3x + 2) + \frac{\sqrt[3]{-2-x}}{x-3}$??? $(-\infty, 1) \cup (2, 3) \cup (3, \infty)$

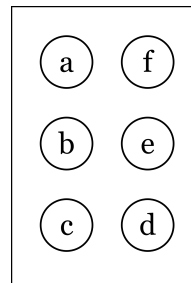
(c) $h(t) = \frac{\sqrt{9-t^2}}{\log_3(2-t)}$??? $\langle -3, 1 \rangle \cup (1, 2)$

(d) $u(x) = \frac{-4x-3}{2x+1}$??? $(-\frac{3}{4}, -\frac{1}{2})$

(e) $\varphi(t) = 3t^3 - 6t^2 + \frac{1}{t} + 4$, $t_0 = -\frac{1}{4}$??? $\frac{213}{64}$

(f) $w(x) = \sqrt[3]{5x^2+3x+2}$??? 1

3.



Braille letter

Function of one variable, group *Sigma* σ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 2 \arccos\left(\frac{7-5x}{3}\right) - \frac{e^{2x+2}}{\sqrt{-30x^2+42x}}$??? $\langle \frac{4}{5}, \frac{7}{5} \rangle$

(b) $g(x) = \frac{1}{2} \ln(x^2 + 11x + 18) + \frac{\sqrt[3]{-7-x}}{x+1}$??? $(-\infty, -9) \cup (-2, -1) \cup (-1, \infty)$

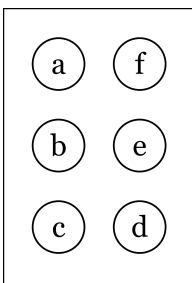
(c) $h(t) = \frac{\sqrt{9-t^2}}{\log_6(1-\frac{1}{2}t)}$??? $\langle -3, 0 \rangle \cup (0, 2)$

(d) $u(x) = \frac{3x+4}{-2x-6}$??? $(-3, -\frac{4}{3})$

(e) $\varphi(t) = 7t^3 - 5t^2 + \frac{3}{t} - 9$, $t_0 = -\frac{1}{4}$??? $-\frac{1371}{64}$

(f) $w(x) = \sqrt[3]{6x^2-5x+8}$??? 2

4.



Braille letter

Function of one variable, group *Tau* τ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arcsin\left(\frac{6-3x}{2}\right) - \frac{e^{-2x-5}}{\sqrt{18x^2-36x}} \quad \dots \quad ??? \quad \dots \quad \left(2, \frac{8}{3}\right)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 - 14x + 45) + \frac{\sqrt[3]{-3-x}}{x-10} \quad ??? \quad (-\infty, 5) \cup (9, 10) \cup (10, \infty)$$

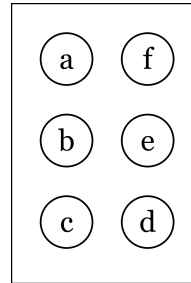
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_4(1+\frac{1}{3}t)} \quad \dots \quad ??? \quad \dots \quad (-3, 0) \cup (0, 4)$$

$$(d) \quad u(x) = \frac{-3x-2}{2x-9} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{-2}{3}, \frac{9}{2} \right\rangle$$

$$(e) \quad \varphi(t) = 2t^3 - t^2 - \frac{6}{t} + 6, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad \frac{17}{2}$$

$$(f) \quad w(x) = \sqrt[3]{-2x^2+6x-5} \quad \dots \quad ??? \quad \dots \quad 2$$

1.



Braille letter

Function of one variable, group *Tau* τ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 2 \arcsin\left(\frac{7-2x}{6}\right) - \frac{e^{3x+6}}{\sqrt{12x^2-42x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{7}{2}, \frac{13}{2}\right)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 + 2x - 48) + \frac{\sqrt[3]{2-x}}{x-7} \quad ??? \quad (-\infty, -8) \cup (6, 7) \cup (7, \infty)$$

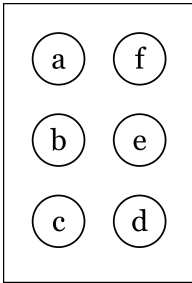
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_4(8+\frac{8}{3}t)} \quad \dots \quad ??? \quad \dots \quad (-3, -\frac{21}{8}) \cup (-\frac{21}{8}, 4)$$

$$(d) \quad u(x) = \frac{-3x+6}{-6x-6} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup (2, \infty)$$

$$(e) \quad \varphi(t) = -t^3 + t^2 + \frac{8}{t} - 1, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{95}{27}$$

$$(f) \quad w(x) = 5x^3 - 23x^2 + 12x \quad \dots \quad ??? \quad \dots \quad 3$$

2.



Braille letter

Function of one variable, group *Tau* τ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arccos\left(\frac{4-4x}{3}\right) - \frac{e^{6x-1}}{\sqrt{-4x^2+4x}} \quad \dots \quad ??? \quad \dots \quad \left(1, \frac{7}{4}\right)$$

$$(b) \quad g(x) = \frac{1}{7} \ln(x^2 + 3x - 4) + \frac{\sqrt[3]{-3-x}}{x-2} \quad ??? \quad (-\infty, -4) \cup (1, 2) \cup (2, \infty)$$

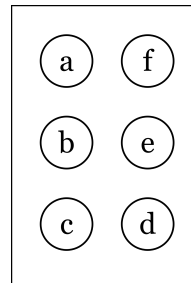
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_5(2+t)} \quad \dots \quad ??? \quad \dots \quad (-2, -1) \cup (-1, 3)$$

$$(d) \quad u(x) = \frac{-4x-1}{3x-1} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{-1}{4}, \frac{1}{3} \right\rangle$$

$$(e) \quad \varphi(t) = -t^3 + 4t^2 - \frac{8}{t} - 7, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{104}{27}$$

$$(f) \quad w(x) = -\frac{x}{3} - \frac{3}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

3.



Braille letter

Function of one variable, group *Tau* τ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arccos\left(\frac{7-4x}{3}\right) - \frac{e^{-5x-7}}{\sqrt{24x^2-42x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{7}{4}, \frac{5}{2}\right)$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 + 12x + 35) + \frac{\sqrt[3]{1-x}}{x+4} \quad ??? \quad (-\infty, -7) \cup (-5, -4) \cup (-4, \infty)$$

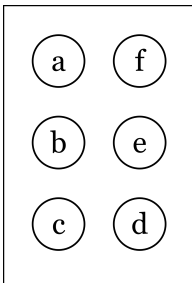
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_6(2-t)} \quad \dots \quad ??? \quad \dots \quad (-3, 1) \cup (1, 2)$$

$$(d) \quad u(x) = \frac{8x+3}{2x-1} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{-3}{8}) \cup (\frac{1}{2}, \infty)$$

$$(e) \quad \varphi(t) = t^3 - 5t^2 - \frac{7}{t} - 1, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{524}{27}$$

$$(f) \quad w(x) = -\frac{x}{7} - \frac{7}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

4.



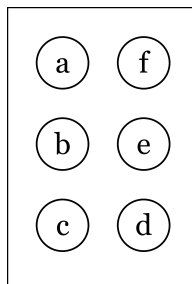
Braille letter

Function of one variable, group *Upsilon v* -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 7 \arccos\left(\frac{5-7x}{3}\right) - \frac{e^{5x+9}}{\sqrt{-56x^2+40x}}$??? $\left\langle \frac{2}{7}, \frac{5}{7} \right\rangle$
 (b) $g(x) = \frac{1}{7} \ln(x^2+3x-4) + \frac{\sqrt[3]{2-x}}{x+5}$??? $(-\infty, -5) \cup (-5, -4) \cup (1, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_6(2-\frac{2}{5}t)}$??? $\left\langle -6, \frac{5}{2} \right\rangle \cup \left(\frac{5}{2}, 5 \right)$
 (d) $u(x) = \frac{-6x-8}{-3x+8}$??? $(-\infty, \frac{-4}{3}) \cup (\frac{8}{3}, \infty)$
 (e) $\varphi(t) = -8t^3 - 2t^2 + \frac{9}{t} - 3$, $t_0 = -\frac{1}{2}$??? -7
 (f) $w(x) = -\frac{x}{4} - \frac{4}{x}$??? 2

1.



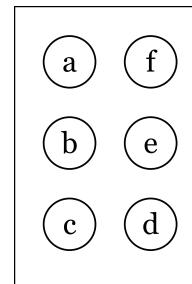
Braille letter

Function of one variable, group *Upsilon v* -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 2 \arcsin\left(\frac{8-8x}{2}\right) - \frac{e^{-8x-1}}{\sqrt{-24x^2+24x}}$... ??? ... $(1, \frac{5}{4})$
 (b) $g(x) = \frac{1}{4} \ln(x^2-9x+14) + \frac{\sqrt[3]{-9-x}}{x-8}$??? $(-\infty, 2) \cup (7, 8) \cup (8, \infty)$
 (c) $h(t) = \frac{\sqrt{25-t^2}}{\log_7(6+\frac{3}{2}t)}$??? $(-4, -\frac{10}{3}) \cup (-\frac{10}{3}, 5)$
 (d) $u(x) = \frac{7x+1}{4x+1}$??? $(-\infty, \frac{-1}{4}) \cup (\frac{-1}{7}, \infty)$
 (e) $\varphi(t) = 2t^3 - 3t^2 - \frac{3}{t} + 9$, $t_0 = -\frac{1}{3}$??? $\frac{259}{27}$
 (f) $w(x) = -x^3 + 7x^2 - 12x$??? 3

2.



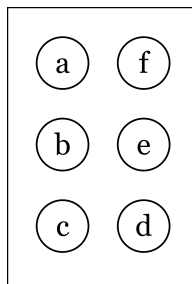
Braille letter

Function of one variable, group *Upsilon v* -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 3 \arccos\left(\frac{4-7x}{2}\right) - \frac{e^{-8x-5}}{\sqrt{7x^2-4x}}$??? $\left\langle \frac{4}{7}, \frac{6}{7} \right\rangle$
 (b) $g(x) = \frac{1}{2} \ln(x^2-6x+8) + \frac{\sqrt[3]{-2-x}}{x-1}$??? $(-\infty, 1) \cup (1, 2) \cup (4, \infty)$
 (c) $h(t) = \frac{\sqrt{36-t^2}}{\log_4(2+\frac{2}{5}t)}$??? $(-5, -\frac{5}{2}) \cup (-\frac{5}{2}, 6)$
 (d) $u(x) = \frac{6x-1}{-6x-4}$??? $(\frac{-2}{3}, \frac{1}{6})$
 (e) $\varphi(t) = 7t^3 + t^2 - \frac{2}{t} + 5$, $t_0 = -\frac{1}{3}$??? $\frac{293}{27}$
 (f) $w(x) = -\frac{x}{9} - \frac{9}{x}$??? 0

3.



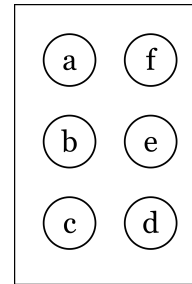
Braille letter

Function of one variable, group *Upsilon v* -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

- (a) $f(x) = 4 \arcsin\left(\frac{6-4x}{2}\right) - \frac{e^{-x+4}}{\sqrt{-8x^2+12x}}$??? $\left\langle 1, \frac{3}{2} \right\rangle$
 (b) $g(x) = \frac{1}{8} \ln(x^2+2x-8) + \frac{\sqrt[3]{-1-x}}{x+5}$??? $(-\infty, -5) \cup (-5, -4) \cup (2, \infty)$
 (c) $h(t) = \frac{\sqrt{4-t^2}}{\log_6(2-2t)}$??? $\left\langle -2, \frac{1}{2} \right\rangle \cup \left(\frac{1}{2}, 1 \right)$
 (d) $u(x) = \frac{3x+5}{-8x-1}$??? $(\frac{-5}{3}, \frac{-1}{8})$
 (e) $\varphi(t) = 4t^3 + 4t^2 - \frac{2}{t} - 3$, $t_0 = -\frac{1}{3}$??? $\frac{89}{27}$
 (f) $w(x) = \sqrt[3]{-3x^2-x-5}$??? 1

4.



Braille letter

Function of one variable, group *Phi* ϕ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -\arccos\left(\frac{4-3x}{2}\right) - \frac{e^{-2x+9}}{\sqrt{12x^2-16x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{4}{3}, 2\right)$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 + 2x - 8) + \frac{\sqrt[3]{-2-x}}{x-3} \quad ??? \quad (-\infty, -4) \cup (2, 3) \cup (3, \infty)$$

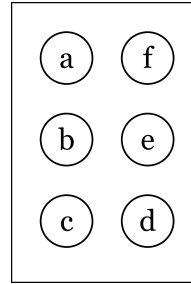
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_4(3+t)} \quad \dots \quad ??? \quad \dots \quad (-3, -2) \cup (-2, 4)$$

$$(d) \quad u(x) = \frac{6x+1}{5x-3} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{-1}{6}) \cup (\frac{3}{5}, \infty)$$

$$(e) \quad \varphi(t) = -6t^3 + 5t^2 + \frac{5}{t} + 3, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{19}{9}$$

$$(f) \quad w(x) = \sqrt[5]{-x^2+3x-4} \quad \dots \quad ??? \quad \dots \quad 0$$

1.



Braille letter

Function of one variable, group *Phi* ϕ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arccos\left(\frac{4-x}{2}\right) - \frac{e^{x-3}}{\sqrt{-5x^2+20x}} \quad \dots \quad ??? \quad \dots \quad (4, 6)$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 + x - 2) + \frac{\sqrt[3]{-9-x}}{x-2} \quad ??? \quad (-\infty, -2) \cup (1, 2) \cup (2, \infty)$$

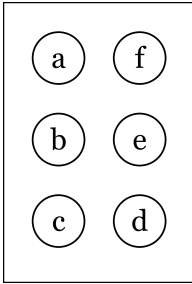
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_8(1+t)} \quad \dots \quad ??? \quad \dots \quad (-1, 0) \cup (0, 2)$$

$$(d) \quad u(x) = \frac{-3x-6}{8x-4} \quad \dots \quad ??? \quad \dots \quad \left\langle -2, \frac{1}{2} \right\rangle$$

$$(e) \quad \varphi(t) = -9t^3 + 7t^2 - \frac{7}{t} + 4, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{67}{9}$$

$$(f) \quad w(x) = -\frac{x}{2} - \frac{2}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

2.



Braille letter

Function of one variable, group *Phi* ϕ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -4 \arccos\left(\frac{7-2x}{2}\right) - \frac{e^{2x-1}}{\sqrt{-4x^2+14x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{5}{2}, \frac{7}{2} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{8} \ln(x^2 + 5x - 14) + \frac{\sqrt[3]{-1-x}}{x-3} \quad ??? \quad (-\infty, -7) \cup (2, 3) \cup (3, \infty)$$

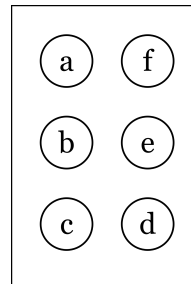
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_2(1+\frac{1}{3}t)} \quad \dots \quad ??? \quad \dots \quad (-3, 0) \cup (0, 4)$$

$$(d) \quad u(x) = \frac{-x-1}{-8x+2} \quad \dots \quad ??? \quad \dots \quad (-\infty, -1) \cup (\frac{1}{4}, \infty)$$

$$(e) \quad \varphi(t) = 7t^3 - 5t^2 + \frac{6}{t} - 4, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{379}{64}$$

$$(f) \quad w(x) = \frac{x}{3} + \frac{3}{x} \quad \dots \quad ??? \quad \dots \quad 1$$

3.



Braille letter

Function of one variable, group *Phi* ϕ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arcsin\left(\frac{4-x}{2}\right) - \frac{e^{-x-4}}{\sqrt{2x^2-8x}} \quad \dots \quad ??? \quad \dots \quad (4, 6)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - x - 6) + \frac{\sqrt[3]{-7-x}}{x+3} \quad ??? \quad (-\infty, -3) \cup (-3, -2) \cup (3, \infty)$$

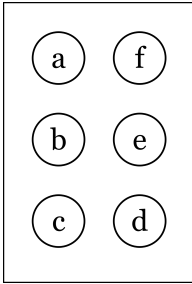
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_8(8+8t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{7}{8}) \cup (-\frac{7}{8}, 2)$$

$$(d) \quad u(x) = \frac{3x-3}{x-8} \quad \dots \quad ??? \quad \dots \quad (-\infty, 1) \cup (8, \infty)$$

$$(e) \quad \varphi(t) = -4t^3 + 2t^2 - \frac{2}{t} - 7, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad \frac{19}{16}$$

$$(f) \quad w(x) = -4x^3 + 5x^2 - x \quad \dots \quad ??? \quad \dots \quad 1$$

4.



Braille letter

Function of one variable, group *Chi* χ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -4 \arccos\left(\frac{7-6x}{2}\right) - \frac{e^{3x-1}}{\sqrt{30x^2-35x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{7}{6}, \frac{3}{2}\right)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 2x - 3) + \frac{\sqrt[3]{-5-x}}{x-4} \quad ??? \quad (-\infty, -1) \cup (3, 4) \cup (4, \infty)$$

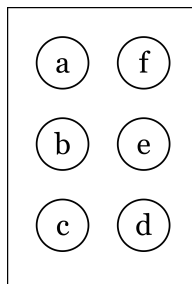
$$(c) \quad h(t) = \frac{\sqrt{16-t^2}}{\log_2(1+\frac{1}{3}t)} \quad \dots \quad ??? \quad \dots \quad (-3, 0) \cup (0, 4)$$

$$(d) \quad u(x) = \frac{-x+3}{-5x+3} \quad \dots \quad ??? \quad \dots \quad (-\infty, \frac{3}{5}) \cup (3, \infty)$$

$$(e) \quad \varphi(t) = -3t^3 + 6t^2 + \frac{3}{t} + 2, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{56}{9}$$

$$(f) \quad w(x) = -\frac{x}{6} - \frac{6}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

1.



Braille letter

Function of one variable, group *Chi* χ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = \arccos\left(\frac{9-x}{6}\right) - \frac{e^{4x-8}}{\sqrt{-7x^2+63x}} \quad \dots \quad ??? \quad \dots \quad (3, 9)$$

$$(b) \quad g(x) = \frac{1}{5} \ln(x^2 - 7x + 12) + \frac{\sqrt[3]{-3-x}}{x-2} \quad ??? \quad (-\infty, 2) \cup (2, 3) \cup (4, \infty)$$

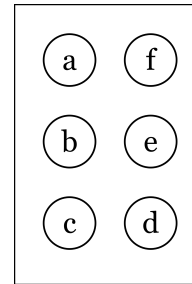
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_2(7+7t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{6}{7}) \cup (-\frac{6}{7}, 2)$$

$$(d) \quad u(x) = \frac{x+5}{3x-1} \quad \dots \quad ??? \quad \dots \quad (-\infty, -5) \cup (\frac{1}{3}, \infty)$$

$$(e) \quad \varphi(t) = -8t^3 + 3t^2 - \frac{6}{t} - 2, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad -\frac{3}{16}$$

$$(f) \quad w(x) = \frac{x}{9} + \frac{9}{x} \quad \dots \quad ??? \quad \dots \quad 2$$

2.



Braille letter

Function of one variable, group *Chi* χ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 6 \arccos\left(\frac{5-8x}{4}\right) - \frac{e^{2x+5}}{\sqrt{72x^2-45x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{1}{8}, \frac{5}{8}\right)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + 14x + 45) + \frac{\sqrt[3]{4-x}}{x+10} \quad ??? \quad (-\infty, -10) \cup (-10, -9) \cup (-5, \infty)$$

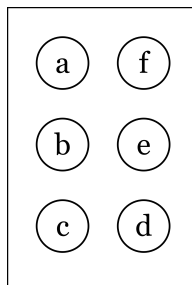
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_4(4+4t)} \quad \dots \quad ??? \quad \dots \quad (-1, -\frac{3}{4}) \cup (-\frac{3}{4}, 2)$$

$$(d) \quad u(x) = \frac{-5x-6}{-7x-2} \quad \dots \quad ??? \quad \dots \quad (-\infty, -\frac{6}{5}) \cup (-\frac{2}{7}, \infty)$$

$$(e) \quad \varphi(t) = -7t^3 - 7t^2 - \frac{1}{t} - 4, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{41}{27}$$

$$(f) \quad w(x) = \frac{x}{3} + \frac{3}{x} \quad \dots \quad ??? \quad \dots \quad 0$$

3.



Braille letter

Function of one variable, group *Chi* χ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -2 \arccos\left(\frac{8-5x}{7}\right) - \frac{e^{-2x-6}}{\sqrt{-40x^2+64x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{1}{5}, \frac{8}{5}\right)$$

$$(b) \quad g(x) = \frac{1}{9} \ln(x^2 - 4x - 12) + \frac{\sqrt[3]{-2-x}}{x-7} \quad ??? \quad (-\infty, -2) \cup (6, 7) \cup (7, \infty)$$

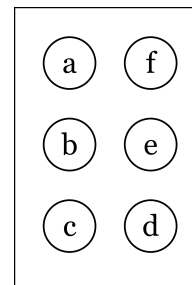
$$(c) \quad h(t) = \frac{\sqrt{25-t^2}}{\log_9(4-t)} \quad \dots \quad ??? \quad \dots \quad (-5, 3) \cup (3, 4)$$

$$(d) \quad u(x) = \frac{-2x+4}{8x+3} \quad \dots \quad ??? \quad \dots \quad (-\frac{3}{8}, 2)$$

$$(e) \quad \varphi(t) = 8t^3 - t^2 - \frac{7}{t} - 9, \quad t_0 = -\frac{1}{2} \quad \dots \quad ??? \quad \dots \quad \frac{15}{4}$$

$$(f) \quad w(x) = -\frac{x}{4} - \frac{4}{x} \quad \dots \quad ??? \quad \dots \quad 1$$

4.



Braille letter

Function of one variable, group *Psi* ψ -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -\arccos\left(\frac{8-2x}{5}\right) - \frac{e^{-3x+2}}{\sqrt{10x^2-40x}}$??? $\left(4, \frac{13}{2}\right)$

(b) $g(x) = \frac{1}{4} \ln(x^2 + 7x + 10) + \frac{\sqrt[3]{1-x}}{x+1}$??? $(-\infty, -5) \cup (-2, -1) \cup (-1, \infty)$

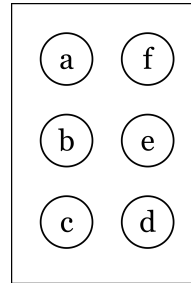
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_8(2+\frac{2}{5}t)}$??? $(-5, -\frac{5}{2}) \cup (-\frac{5}{2}, 6)$

(d) $u(x) = \frac{-8x+4}{-8x-5}$??? $(-\infty, -\frac{5}{8}) \cup (\frac{1}{2}, \infty)$

(e) $\varphi(t) = 3t^3 - t^2 + \frac{5}{t} - 1$, $t_0 = -\frac{1}{3}$??? $-\frac{26}{9}$

(f) $w(x) = \sqrt[3]{-4x^2-5x-4}$??? 0

1.



Braille letter

Function of one variable, group *Psi* ψ -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 7 \arcsin\left(\frac{4-7x}{2}\right) - \frac{e^{2x-6}}{\sqrt{7x^2-4x}}$??? $\left(\frac{4}{7}, \frac{6}{7}\right)$

(b) $g(x) = \frac{1}{4} \ln(x^2 + 5x + 6) + \frac{\sqrt[3]{4-x}}{x+1}$??? $(-\infty, -3) \cup (-2, -1) \cup (-1, \infty)$

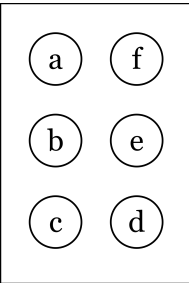
(c) $h(t) = \frac{\sqrt{81-t^2}}{\log_5(9-\frac{9}{8}t)}$??? $\langle -9, \frac{64}{9} \rangle \cup (\frac{64}{9}, 8)$

(d) $u(x) = \frac{-9x+7}{-x+1}$??? $(-\infty, \frac{7}{9}) \cup (1, \infty)$

(e) $\varphi(t) = -9t^3 - t^2 - \frac{3}{t} + 7$, $t_0 = -\frac{1}{3}$??? $\frac{146}{9}$

(f) $w(x) = \sqrt[5]{-3x^2+3x-9}$??? 1

2.



Braille letter

Function of one variable, group *Psi* ψ -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = 8 \arccos\left(\frac{4-6x}{3}\right) - \frac{e^{x+9}}{\sqrt{54x^2-36x}}$??? $\left(\frac{2}{3}, \frac{7}{6}\right)$

(b) $g(x) = \frac{1}{9} \ln(x^2 + 3x - 18) + \frac{\sqrt[3]{-3-x}}{x+7}$??? $(-\infty, -7) \cup (-7, -6) \cup (3, \infty)$

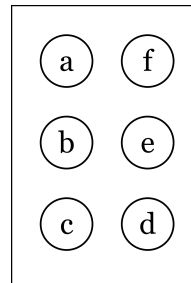
(c) $h(t) = \frac{\sqrt{9-t^2}}{\log_6(4+2t)}$??? $(-2, -\frac{3}{2}) \cup (-\frac{3}{2}, 3)$

(d) $u(x) = \frac{-4x+2}{-5x+1}$??? $(-\infty, \frac{1}{5}) \cup (\frac{1}{2}, \infty)$

(e) $\varphi(t) = 6t^3 + 3t^2 - \frac{3}{t} + 1$, $t_0 = -\frac{1}{2}$??? $\frac{5}{2}$

(f) $w(x) = \frac{x}{4} + \frac{4}{x}$??? 1

3.



Braille letter

Function of one variable, group *Psi* ψ -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

(a) $f(x) = -5 \arccos\left(\frac{6-9x}{2}\right) - \frac{e^{5x-2}}{\sqrt{-27x^2+18x}}$... ??? ... $\langle \frac{4}{9}, \frac{2}{3} \rangle$

(b) $g(x) = \frac{1}{3} \ln(x^2 - x - 12) + \frac{\sqrt[3]{4-x}}{x+4}$??? $(-\infty, -4) \cup (-4, -3) \cup (4, \infty)$

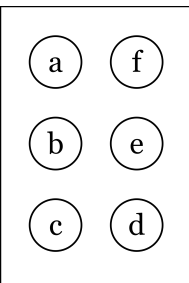
(c) $h(t) = \frac{\sqrt{36-t^2}}{\log_2(2-\frac{2}{5}t)}$??? $\langle -6, \frac{5}{2} \rangle \cup (\frac{5}{2}, 5)$

(d) $u(x) = \frac{3x+5}{-3x+4}$??? $\langle \frac{-5}{3}, \frac{4}{3} \rangle$

(e) $\varphi(t) = -8t^3 - 6t^2 - \frac{2}{t} + 6$, $t_0 = -\frac{1}{2}$??? $\frac{19}{2}$

(f) $w(x) = 5x^3 + 14x^2 + 9x$??? 3

4.



Braille letter

Function of one variable, group *Omega* ω -i

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -5 \arccos\left(\frac{6-6x}{2}\right) - \frac{e^{4x+2}}{\sqrt{-42x^2 + 42x}} \quad \dots \quad ??? \quad \dots \quad \left(1, \frac{4}{3}\right)$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 + 3x - 10) + \frac{\sqrt[3]{-3-x}}{x+6} \quad ??? \quad (-\infty, -6) \cup (-6, -5) \cup (2, \infty)$$

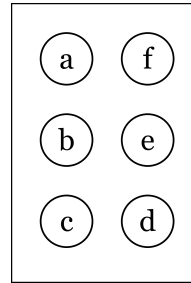
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_3(5-\frac{5}{2}t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -3, \frac{8}{5} \right\rangle \cup \left(\frac{8}{5}, 2 \right)$$

$$(d) \quad u(x) = \frac{-x-3}{6x+2} \quad \dots \quad ??? \quad \dots \quad \left(-3, \frac{-1}{3}\right)$$

$$(e) \quad \varphi(t) = -t^3 + 4t^2 + \frac{4}{t} - 6, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad -\frac{473}{27}$$

$$(f) \quad w(x) = -5x^3 + 31x^2 - 30x \quad \dots \quad ??? \quad \dots \quad 3$$

1.



Braille letter

Function of one variable, group *Omega* ω -ii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = -3 \arcsin\left(\frac{9-4x}{3}\right) - \frac{e^{-3x+4}}{\sqrt{4x^2 - 9x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{9}{4}, 3\right)$$

$$(b) \quad g(x) = \frac{1}{2} \ln(x^2 - 8x + 12) + \frac{\sqrt[3]{6-x}}{x-1} \quad ??? \quad (-\infty, 1) \cup (1, 2) \cup (6, \infty)$$

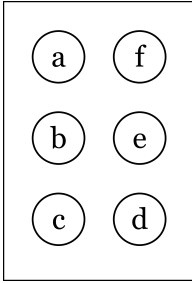
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_6(2+t)} \quad \dots \quad ??? \quad \dots \quad (-2, -1) \cup (-1, 3)$$

$$(d) \quad u(x) = \frac{-2x+1}{-6x-3} \quad \dots \quad ??? \quad \dots \quad \left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$$

$$(e) \quad \varphi(t) = 8t^3 + t^2 - \frac{4}{t} + 4, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{427}{27}$$

$$(f) \quad w(x) = -5x^3 + 4x^2 + x \quad \dots \quad ??? \quad \dots \quad 1$$

2.



Braille letter

Function of one variable, group *Omega* ω -iii

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 4 \arccos\left(\frac{3-8x}{2}\right) - \frac{e^{3x-1}}{\sqrt{24x^2 - 9x}} \quad \dots \quad ??? \quad \dots \quad \left\langle \frac{1}{8}, \frac{3}{8} \right\rangle$$

$$(b) \quad g(x) = \frac{1}{4} \ln(x^2 + 5x - 6) + \frac{\sqrt[3]{-2-x}}{x+7} \quad ??? \quad (-\infty, -7) \cup (-7, -6) \cup (1, \infty)$$

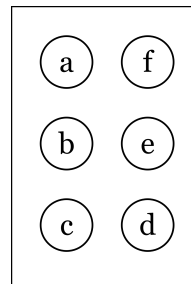
$$(c) \quad h(t) = \frac{\sqrt{9-t^2}}{\log_3(7-\frac{7}{2}t)} \quad \dots \quad ??? \quad \dots \quad \left\langle -3, \frac{12}{7} \right\rangle \cup \left(\frac{12}{7}, 2 \right)$$

$$(d) \quad u(x) = \frac{x+1}{-4x+1} \quad \dots \quad ??? \quad \dots \quad \left\langle -1, \frac{1}{4} \right\rangle$$

$$(e) \quad \varphi(t) = -5t^3 - 3t^2 - \frac{1}{t} + 9, \quad t_0 = -\frac{1}{3} \quad \dots \quad ??? \quad \dots \quad \frac{248}{27}$$

$$(f) \quad w(x) = -x^3 - 6x^2 - 8x \quad \dots \quad ??? \quad \dots \quad 3$$

3.



Braille letter

Function of one variable, group *Omega* ω -iv

- (a) Determine the domain of the function $f(x)$. (e) Determine the value of the function $\varphi(t)$ at the point t_0 .
 (b) Determine the domain of the function $g(x)$. (f) Determine the number of intersection points of the function plot $w(x)$ with the x -axis.
 (c) Determine the domain of the function $h(t)$.
 (d) Determine the interval on which the function $u(x)$ is non-negative.

$$(a) \quad f(x) = 2 \arccos\left(\frac{5-2x}{2}\right) - \frac{e^{-2x-1}}{\sqrt{6x^2 - 15x}} \quad \dots \quad ??? \quad \dots \quad \left(\frac{5}{2}, \frac{7}{2}\right)$$

$$(b) \quad g(x) = \frac{1}{3} \ln(x^2 + 8x + 7) + \frac{\sqrt[3]{4-x}}{x+8} \quad ??? \quad (-\infty, -8) \cup (-8, -7) \cup (-1, \infty)$$

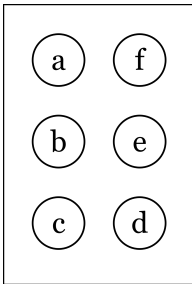
$$(c) \quad h(t) = \frac{\sqrt{4-t^2}}{\log_6(8+8t)} \quad \dots \quad ??? \quad \dots \quad \left(-1, -\frac{7}{8}\right) \cup \left(-\frac{7}{8}, 2\right)$$

$$(d) \quad u(x) = \frac{-5x-1}{3x+1} \quad \dots \quad ??? \quad \dots \quad \left(\frac{-1}{3}, \frac{-1}{5}\right)$$

$$(e) \quad \varphi(t) = -8t^3 + t^2 + \frac{1}{t} + 4, \quad t_0 = -\frac{1}{4} \quad \dots \quad ??? \quad \dots \quad \frac{63}{16}$$

$$(f) \quad w(x) = -x^3 + 3x^2 + 28x \quad \dots \quad ??? \quad \dots \quad 3$$

4.



Braille letter

Function of one variable (solutions)

α	$i : \mathbf{B}$	(a) $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (1, 2) \cup (5, \infty) \checkmark$	(c) $(-3, 0) \cup (0, 4) \times$	(d) $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{5}{2}, \infty\right) \times$	(e) $\frac{145}{64} \times$	(f) 3 $\times$
	$ii : \mathbf{I}$	(a) $\left\langle \frac{5}{4}, 2 \right\rangle \times$	(b) $(-\infty, -5) \cup (6, 7) \cup (7, \infty) \checkmark$	(c) $\left(-8, -\frac{48}{7}\right) \cup \left(-\frac{48}{7}, 9\right) \times$	(d) $\left(\frac{-1}{2}, 3\right) \times$	(e) $-\frac{41}{9} \times$	(f) 0 $\checkmark$
	$iii : \mathbf{R}$	(a) $(3, 4) \checkmark$	(b) $(-\infty, 3) \cup (9, 10) \cup (10, \infty) \checkmark$	(c) $\left\langle -2, \frac{2}{3} \right\rangle \cup \left(\frac{2}{3}, 1\right) \checkmark$	(d) $\left\langle \frac{-5}{8}, \frac{-1}{5} \right\rangle \times$	(e) $-\frac{43}{8} \checkmark$	(f) 3 $\times$
	$iv : \mathbf{D}$	(a) $(1, 2) \checkmark$	(b) $(-\infty, -8) \cup (-3, -2) \cup (-2, \infty) \times$	(c) $(-6, -3) \cup (-3, 7) \times$	(d) $\left\langle \frac{-9}{2}, -\frac{2}{3} \right\rangle \times$	(e) $\frac{29}{2} \checkmark$	(f) 0 $\checkmark$
β	$i : \mathbf{F}$	(a) $\left\langle \frac{6}{5}, 2 \right\rangle \checkmark$	(b) $(-\infty, -3) \cup (-2, -1) \cup (-1, \infty) \checkmark$	(c) $(-4, 0) \cup (0, 3) \times$	(d) $\left(-\infty, \frac{4}{3}\right) \cup (2, \infty) \times$	(e) $-\frac{43}{8} \times$	(f) 0 $\checkmark$
	$ii : \mathbf{A}$	(a) $(1, 3) \checkmark$	(b) $(-\infty, 5) \cup (5, 6) \cup (8, \infty) \times$	(c) $(-7, 5) \cup (5, 6) \times$	(d) $(-\infty, -1) \cup \left(\frac{-1}{5}, \infty\right) \times$	(e) $-\frac{97}{8} \times$	(f) 3 $\times$
	$iii : \mathbf{R}$	(a) $\left\langle \frac{4}{3}, 2 \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (-8, -7) \cup (-5, \infty) \checkmark$	(c) $(-3, 1) \cup (1, 2) \checkmark$	(d) $(-8, 5) \times$	(e) $-\frac{29}{2} \checkmark$	(f) 0 $\times$
	$iv : \mathbf{M}$	(a) $(1, 3) \checkmark$	(b) $(-\infty, -3) \cup (2, 3) \cup (3, \infty) \times$	(c) $\left\langle -5, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 4\right) \checkmark$	(d) $\left(-\infty, -\frac{2}{5}\right) \cup \left(\frac{-1}{4}, \infty\right) \times$	(e) $-\frac{35}{27} \times$	(f) 0 $\checkmark$
γ	$i : \mathbf{T}$	(a) $(8, 13) \times$	(b) $(-\infty, -1) \cup (4, 5) \cup (5, \infty) \checkmark$	(c) $\left(-1, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 2\right) \checkmark$	(d) $(-\infty, -2) \cup \left(\frac{1}{5}, \infty\right) \times$	(e) $-\frac{61}{8} \checkmark$	(f) 0 $\checkmark$
	$ii : \mathbf{E}$	(a) $\left\langle \frac{7}{4}, \frac{9}{4} \right\rangle \checkmark$	(b) $(-\infty, 4) \cup (4, 5) \cup (6, \infty) \times$	(c) $(-2, -1) \cup (-1, 3) \times$	(d) $\left(-6, -\frac{1}{3}\right) \times$	(e) $-\frac{27}{8} \checkmark$	(f) 0 $\times$
	$iii : \mathbf{S}$	(a) $\left\langle 4, \frac{11}{2} \right\rangle \times$	(b) $(-\infty, -7) \cup (-5, -4) \cup (-4, \infty) \checkmark$	(c) $(-2, 0) \cup (0, 1) \checkmark$	(d) $\left(-\frac{4}{7}, \frac{1}{2}\right) \times$	(e) $-\frac{251}{9} \times$	(f) 0 $\checkmark$
	$iv : \mathbf{T}$	(a) $\left\langle \frac{1}{3}, \frac{2}{3} \right\rangle \times$	(b) $(-\infty, -3) \cup (5, 6) \cup (6, \infty) \checkmark$	(c) $\left\langle -2, \frac{6}{7} \right\rangle \cup \left(\frac{6}{7}, 1\right) \checkmark$	(d) $(-\infty, -4) \cup \left(\frac{-3}{4}, \infty\right) \times$	(e) $-\frac{489}{16} \checkmark$	(f) 0 $\checkmark$
δ	$i : \mathbf{A}$	(a) $\left\langle 3, \frac{13}{2} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (4, 5) \cup (5, \infty) \times$	(c) $(-6, 0) \cup (0, 7) \times$	(d) $\left(-\infty, \frac{3}{7}\right) \cup (3, \infty) \times$	(e) $\frac{29}{2} \times$	(f) 0 $\times$
	$ii : \mathbf{R}$	(a) $\left\langle \frac{4}{5}, \frac{6}{5} \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (7, 8) \cup (8, \infty) \checkmark$	(c) $\left(-3, -\frac{3}{2}\right) \cup \left(-\frac{3}{2}, 4\right) \checkmark$	(d) $(-\infty, -4) \cup (-1, \infty) \times$	(e) $\frac{1}{2} \checkmark$	(f) 0 $\times$
	$iii : \mathbf{M}$	(a) $(2, 3) \checkmark$	(b) $(-\infty, -5) \cup (-5, -4) \cup (-3, \infty) \times$	(c) $\left(-1, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 2\right) \checkmark$	(d) $\left(-\infty, \frac{-7}{4}\right) \cup (1, \infty) \times$	(e) $\frac{1025}{64} \times$	(f) 3 $\checkmark$
	$iv : \mathbf{Y}$	(a) $\left\langle \frac{3}{4}, \frac{5}{4} \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (-5, -4) \cup (-2, \infty) \times$	(c) $\left(-8, -\frac{32}{5}\right) \cup \left(-\frac{32}{5}, 9\right) \checkmark$	(d) $\left(\frac{2}{3}, -1\right) \checkmark$	(e) $\frac{217}{32} \checkmark$	(f) 0 $\checkmark$
ϵ	$i : \mathbf{E}$	(a) $\left\langle \frac{1}{5}, \frac{3}{5} \right\rangle \checkmark$	(b) $(-\infty, -3) \cup (8, 9) \cup (9, \infty) \times$	(c) $\left(-5, -\frac{10}{3}\right) \cup \left(-\frac{10}{3}, 6\right) \times$	(d) $\left(-\infty, \frac{2}{5}\right) \cup \left(\frac{3}{5}, \infty\right) \times$	(e) $-\frac{71}{8} \checkmark$	(f) 0 $\times$
	$ii : \mathbf{A}$	(a) $\left\langle \frac{9}{7}, \frac{13}{7} \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (2, 3) \cup (3, \infty) \times$	(c) $(-3, 1) \cup (1, 2) \times$	(d) $(-\infty, -6) \cup \left(\frac{2}{3}, \infty\right) \times$	(e) $-\frac{121}{8} \times$	(f) 0 $\times$
	$iii : \mathbf{S}$	(a) $(4, 6) \times$	(b) $(-\infty, -2) \cup (8, 9) \cup (9, \infty) \checkmark$	(c) $\left(-2, -\frac{8}{5}\right) \cup \left(-\frac{8}{5}, 3\right) \checkmark$	(d) $\left\langle \frac{-7}{8}, \frac{-1}{9} \right\rangle \times$	(e) $-\frac{124}{27} \times$	(f) 0 $\checkmark$
	$iv : \mathbf{T}$	(a) $\left\langle 2, \frac{3}{2} \right\rangle \times$	(b) $(-\infty, -10) \cup (-10, -9) \cup (6, \infty) \checkmark$	(c) $(-2, 0) \cup (0, 3) \checkmark$	(d) $(-\infty, -1) \cup \left(\frac{1}{2}, \infty\right) \times$	(e) $-\frac{860}{27} \checkmark$	(f) 0 $\checkmark$
ζ	$i : \mathbf{T}$	(a) $\left\langle \frac{3}{8}, \frac{5}{8} \right\rangle \times$	(b) $(-\infty, -5) \cup (7, 8) \cup (8, \infty) \checkmark$	(c) $\left\langle -3, \frac{4}{3} \right\rangle \cup \left(\frac{4}{3}, 2\right) \checkmark$	(d) $\left(\frac{3}{2}, \frac{7}{2}\right) \times$	(e) $\frac{770}{27} \checkmark$	(f) 3 $\checkmark$
	$ii : \mathbf{E}$	(a) $(9, 13) \checkmark$	(b) $(-\infty, -9) \cup (-7, -6) \cup (-6, \infty) \times$	(c) $(-4, 0) \cup (0, 5) \times$	(d) $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{4}{5}, \infty\right) \times$	(e) $-\frac{35}{8} \checkmark$	(f) 0 $\times$
	$iii : \mathbf{A}$	(a) $\left\langle \frac{7}{5}, 2 \right\rangle \checkmark$	(b) $(-\infty, -3) \cup (2, 3) \cup (3, \infty) \times$	(c) $(-2, -1) \cup (-1, 3) \times$	(d) $\left(-\frac{1}{6}, \frac{1}{3}\right) \times$	(e) $-\frac{187}{27} \times$	(f) 3 $\times$
	$iv : \mathbf{M}$	(a) $\left\langle \frac{1}{8}, \frac{3}{4} \right\rangle \checkmark$	(b) $(-\infty, 3) \cup (3, 4) \cup (5, \infty) \times$	(c) $\left(-1, -\frac{4}{5}\right) \cup \left(-\frac{4}{5}, 2\right) \checkmark$	(d) $\left(-\infty, \frac{7}{3}\right) \cup (4, \infty) \times$	(e) $\frac{151}{27} \times$	(f) 3 $\checkmark$
η	$i : \mathbf{S}$	(a) $(6, 9) \times$	(b) $(-\infty, -5) \cup (-5, -4) \cup (-1, \infty) \checkmark$	(c) $\left(-2, -\frac{12}{7}\right) \cup \left(-\frac{12}{7}, 3\right) \checkmark$	(d) $\left\langle 2, \frac{5}{2} \right\rangle \times$	(e) $-\frac{7}{2} \times$	(f) 3 $\checkmark$
	$ii : \mathbf{H}$	(a) $\left\langle 1, \frac{3}{2} \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (2, 3) \cup (3, \infty) \checkmark$	(c) $(-5, -4) \cup (-4, 6) \times$	(d) $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{5}, \infty\right) \times$	(e) $-\frac{143}{32} \checkmark$	(f) 0 $\times$
	$iii : \mathbf{O}$	(a) $\left\langle \frac{4}{3}, 3 \right\rangle \checkmark$	(b) $(-\infty, -2) \cup (3, 4) \cup (4, \infty) \times$	(c) $\left\langle -4, \frac{5}{2} \right\rangle \cup \left(\frac{5}{2}, 3\right) \checkmark$	(d) $(-\infty, -1) \cup \left(\frac{1}{7}, \infty\right) \times$	(e) $\frac{929}{64} \checkmark$	(f) 3 $\times$
	$iv : \mathbf{P}$	(a) $\left\langle \frac{1}{3}, \frac{2}{3} \right\rangle \checkmark$	(b) $(-\infty, 3) \cup (6, 7) \cup (7, \infty) \checkmark$	(c) $(-2, 0) \cup (0, 3) \checkmark$	(d) $(-\infty, -1) \cup \left(\frac{5}{2}, \infty\right) \times$	(e) $\frac{365}{27} \times$	(f) 3 $\checkmark$
θ	$i : \mathbf{L}$	(a) $(2, 3) \checkmark$	(b) $(-\infty, 3) \cup (7, 8) \cup (8, \infty) \checkmark$	(c) $\left(-1, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 2\right) \checkmark$	(d) $\left(-\infty, \frac{1}{4}\right) \cup \left(\frac{5}{4}, \infty\right) \times$	(e) $-\frac{57}{4} \times$	(f) 0 $\times$
	$ii : \mathbf{A}$	(a) $\left\langle \frac{1}{5}, 1 \right\rangle \checkmark$	(b) $(-\infty, -7) \cup (-7, -6) \cup (5, \infty) \times$	(c) $\left\langle -2, \frac{6}{7} \right\rangle \cup \left(\frac{6}{7}, 1\right) \times$	(d) $\left(-\infty, \frac{6}{5}\right) \cup \left(\frac{5}{3}, \infty\right) \times$	(e) $-\frac{293}{16} \times$	(f) 0 $\times$
	$iii : \mathbf{N}$	(a) $\left\langle \frac{7}{3}, 4 \right\rangle \checkmark$	(b) $(-\infty, 2) \cup (2, 3) \cup (9, \infty) \times$	(c) $(-2, 0) \cup (0, 3) \checkmark$	(d) $\left\langle -2, -\frac{1}{3} \right\rangle \times$	(e) $-\frac{57}{8} \checkmark$	(f) 0 $\checkmark$
	$iv : \mathbf{D}$	(a) $\left\langle \frac{6}{5}, \frac{8}{5} \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (-4, -3) \cup (4, \infty) \times$	(c) $\left\langle -4, \frac{18}{7} \right\rangle \cup \left(\frac{18}{7}, 3\right) \times$	(d) $\left(-\infty, \frac{1}{7}\right) \cup \left(\frac{7}{2}, \infty\right) \times$	(e) $-\frac{269}{27} \checkmark$	(f) 3 $\checkmark$

Function of one variable (solutions)

$\boxed{\mu}$	$i : \mathbf{F}$	(a) $\left\langle \frac{1}{9}, \frac{2}{3} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (1, 2) \cup (5, \infty) \checkmark$	(c) $\langle -5, 3 \rangle \cup (3, 4) \times$	(d) $\left\langle -\frac{5}{4}, \frac{2}{3} \right\rangle \times$	(e) $-\frac{467}{27} \times$	(f) $3 \checkmark$
	$ii : \mathbf{O}$	(a) $\left\langle \frac{7}{3}, \frac{11}{3} \right\rangle \checkmark$	(b) $(-\infty, 4) \cup (5, 6) \cup (6, \infty) \times$	(c) $\left(-3, -\frac{8}{3}\right) \cup \left(-\frac{8}{3}, 4\right) \checkmark$	(d) $\left(-1, \frac{1}{6}\right) \times$	(e) $-\frac{1039}{32} \checkmark$	(f) $0 \times$
	$iii : \mathbf{O}$	(a) $\left\langle \frac{4}{3}, \frac{8}{3} \right\rangle \checkmark$	(b) $(-\infty, -6) \cup (-6, -5) \cup (1, \infty) \times$	(c) $\left(-1, -\frac{3}{4}\right) \cup \left(-\frac{3}{4}, 2\right) \checkmark$	(d) $\left(-\infty, -\frac{4}{5}\right) \cup \left(-\frac{4}{5}, \infty\right) \times$	(e) $-\frac{87}{8} \checkmark$	(f) $3 \times$
	$iv : \mathbf{T}$	(a) $\langle 8, 10 \rangle \times$	(b) $(-\infty, -8) \cup (-8, -7) \cup (2, \infty) \checkmark$	(c) $\left\langle -4, \frac{3}{2} \right\rangle \cup \left(\frac{3}{2}, 3\right) \checkmark$	(d) $\left(-\infty, \frac{5}{2}\right) \cup \left\langle \frac{5}{2}, \infty\right\rangle \times$	(e) $-\frac{176}{27} \checkmark$	(f) $0 \checkmark$
$\boxed{\kappa}$	$i : \mathbf{C}$	(a) $\left\langle \frac{5}{8}, \frac{9}{8} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (6, 7) \cup (7, \infty) \times$	(c) $\left\langle -5, \frac{10}{3} \right\rangle \cup \left(\frac{10}{3}, 4\right) \times$	(d) $(-\infty, -8) \cup (-1, \infty) \times$	(e) $-\frac{157}{8} \times$	(f) $0 \checkmark$
	$ii : \mathbf{A}$	(a) $\left\langle \frac{7}{4}, \frac{5}{2} \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (-8, -7) \cup (3, \infty) \times$	(c) $\left\langle -8, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 7\right) \times$	(d) $\left\langle -\frac{6}{5}, \frac{2}{3} \right\rangle \times$	(e) $-\frac{97}{8} \times$	(f) $0 \times$
	$iii : \mathbf{S}$	(a) $\left\langle \frac{1}{4}, 1 \right\rangle \times$	(b) $(-\infty, 1) \cup (4, 5) \cup (5, \infty) \checkmark$	(c) $\left(-5, -\frac{10}{3}\right) \cup \left(-\frac{10}{3}, 6\right) \checkmark$	(d) $(-\infty, -2) \cup (5, \infty) \times$	(e) $-\frac{75}{4} \times$	(f) $3 \checkmark$
	$iv : \mathbf{H}$	(a) $\left\langle \frac{7}{3}, \frac{10}{3} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (8, 9) \cup (9, \infty) \checkmark$	(c) $\left\langle -8, \frac{8}{5} \right\rangle \cup \left(\frac{8}{5}, 2\right) \times$	(d) $\left(-\infty, -\frac{9}{5}\right) \cup \left\langle \frac{1}{3}, \infty\right\rangle \times$	(e) $-\frac{47}{8} \checkmark$	(f) $3 \times$
$\boxed{\lambda}$	$i : \mathbf{L}$	(a) $\left\langle \frac{5}{7}, 1 \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (3, 4) \cup (4, \infty) \checkmark$	(c) $\left\langle -4, \frac{5}{2} \right\rangle \cup \left(\frac{5}{2}, 3\right) \checkmark$	(d) $(-\infty, -2) \cup \left\langle -\frac{1}{2}, \infty\right\rangle \times$	(e) $-\frac{847}{27} \times$	(f) $0 \times$
	$ii : \mathbf{E}$	(a) $\left\langle \frac{3}{8}, \frac{9}{8} \right\rangle \checkmark$	(b) $(-\infty, -2) \cup (6, 7) \cup (7, \infty) \times$	(c) $\left(-2, -\frac{12}{7}\right) \cup \left(-\frac{12}{7}, 3\right) \times$	(d) $\left(-\frac{3}{7}, \frac{1}{5}\right) \times$	(e) $-\frac{1299}{64} \checkmark$	(f) $0 \times$
	$iii : \mathbf{A}$	(a) $\left\langle \frac{3}{2}, 4 \right\rangle \checkmark$	(b) $(-\infty, -9) \cup (-9, -8) \cup (-5, \infty) \times$	(c) $\left\langle -5, \frac{7}{2} \right\rangle \cup \left(\frac{7}{2}, 4\right) \times$	(d) $\left(-\infty, \frac{1}{3}\right) \cup \left\langle \frac{2}{5}, \infty\right\rangle \times$	(e) $-\frac{403}{27} \times$	(f) $3 \times$
	$iv : \mathbf{F}$	(a) $\left\langle \frac{1}{2}, \frac{7}{8} \right\rangle \checkmark$	(b) $(-\infty, 5) \cup (5, 6) \cup (7, \infty) \checkmark$	(c) $\langle -7, 3 \rangle \cup (3, 6) \times$	(d) $\langle -2, 3 \rangle \times$	(e) $-\frac{57}{8} \times$	(f) $0 \checkmark$
$\boxed{\mu}$	$i : \mathbf{P}$	(a) $\langle 7, 9 \rangle \checkmark$	(b) $(-\infty, -7) \cup (-7, -6) \cup (-3, \infty) \checkmark$	(c) $\langle -4, 0 \rangle \cup (0, 5) \checkmark$	(d) $(-\infty, -4) \cup \left\langle -\frac{1}{3}, \infty\right\rangle \times$	(e) $-\frac{353}{27} \times$	(f) $3 \checkmark$
	$ii : \mathbf{A}$	(a) $\left\langle \frac{6}{5}, 2 \right\rangle \checkmark$	(b) $(-\infty, -6) \cup (4, 5) \cup (5, \infty) \times$	(c) $\left(-5, -\frac{10}{3}\right) \cup \left(-\frac{10}{3}, 6\right) \times$	(d) $\left\langle \frac{1}{3}, 1 \right\rangle \times$	(e) $-\frac{15}{4} \times$	(f) $3 \times$
	$iii : \mathbf{T}$	(a) $\left\langle \frac{2}{3}, 2 \right\rangle \times$	(b) $(-\infty, -8) \cup (-8, -7) \cup (-6, \infty) \checkmark$	(c) $\left(-5, -\frac{5}{2}\right) \cup \left(-\frac{5}{2}, 6\right) \checkmark$	(d) $\left(-\infty, -\frac{3}{5}\right) \cup \left(\frac{1}{4}, \infty\right) \times$	(e) $-\frac{109}{8} \checkmark$	(f) $3 \checkmark$
	$iv : \mathbf{H}$	(a) $\left\langle \frac{4}{5}, \frac{6}{5} \right\rangle \checkmark$	(b) $(-\infty, -6) \cup (-6, -5) \cup (-3, \infty) \checkmark$	(c) $\langle -6, 0 \rangle \cup (0, 5) \times$	(d) $\left(-4, -\frac{7}{4}\right) \times$	(e) $-\frac{137}{27} \checkmark$	(f) $3 \times$
$\boxed{\nu}$	$i : \mathbf{A}$	(a) $\left\langle \frac{8}{7}, 2 \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (2, 3) \cup (3, \infty) \times$	(c) $\left(-5, -\frac{15}{4}\right) \cup \left(-\frac{15}{4}, 6\right) \times$	(d) $\left(-\infty, \frac{1}{5}\right) \cup \left\langle \frac{1}{2}, \infty\right\rangle \times$	(e) $-\frac{99}{8} \times$	(f) $3 \times$
	$ii : \mathbf{T}$	(a) $\left\langle \frac{8}{3}, \frac{10}{3} \right\rangle \times$	(b) $(-\infty, -4) \cup (-4, -3) \cup (-1, \infty) \checkmark$	(c) $\langle -5, 0 \rangle \cup (0, 4) \checkmark$	(d) $\left(-\infty, -\frac{1}{6}\right) \cup \left\langle \frac{1}{5}, \infty\right\rangle \times$	(e) $-\frac{2}{3} \checkmark$	(f) $0 \checkmark$
	$iii : \mathbf{O}$	(a) $\left\langle \frac{1}{2}, \frac{3}{2} \right\rangle \checkmark$	(b) $(-\infty, 5) \cup (6, 7) \cup (7, \infty) \times$	(c) $\left\langle -2, \frac{7}{8} \right\rangle \cup \left(\frac{7}{8}, 1\right) \checkmark$	(d) $\left(-\frac{4}{7}, \frac{5}{8}\right) \times$	(e) $-\frac{38}{3} \checkmark$	(f) $0 \times$
	$iv : \mathbf{M}$	(a) $\left\langle \frac{4}{5}, \frac{6}{5} \right\rangle \checkmark$	(b) $(-\infty, -2) \cup (1, 2) \cup (2, \infty) \times$	(c) $\left\langle -3, \frac{16}{9} \right\rangle \cup \left(\frac{16}{9}, 2\right) \checkmark$	(d) $\left(-\infty, -\frac{5}{6}\right) \cup \left\langle \frac{1}{2}, \infty\right\rangle \times$	(e) $-\frac{525}{32} \times$	(f) $0 \checkmark$
$\boxed{\xi}$	$i : \mathbf{G}$	(a) $\left\langle \frac{3}{7}, \frac{5}{7} \right\rangle \checkmark$	(b) $(-\infty, 3) \cup (5, 6) \cup (6, \infty) \checkmark$	(c) $\left(-1, -\frac{5}{6}\right) \cup \left(-\frac{5}{6}, 2\right) \times$	(d) $\langle -8, 3 \rangle \times$	(e) $-\frac{101}{8} \checkmark$	(f) $0 \checkmark$
	$ii : \mathbf{O}$	(a) $\langle 5, 7 \rangle \checkmark$	(b) $(-\infty, -8) \cup (-8, -7) \cup (6, \infty) \times$	(c) $\left(-3, \frac{8}{5}\right) \cup \left(\frac{8}{5}, 2\right) \checkmark$	(d) $(-\infty, -4) \cup \left\langle \frac{5}{2}, \infty\right\rangle \times$	(e) $-\frac{394}{27} \checkmark$	(f) $0 \times$
	$iii : \mathbf{L}$	(a) $\langle 9, 17 \rangle \checkmark$	(b) $(-\infty, -4) \cup (1, 2) \cup (2, \infty) \checkmark$	(c) $\langle -6, 0 \rangle \cup (0, 7) \checkmark$	(d) $(-\infty, -2) \cup (5, \infty) \times$	(e) $-\frac{3}{2} \times$	(f) $0 \times$
	$iv : \mathbf{D}$	(a) $\left\langle \frac{9}{2}, 7 \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (-5, -4) \cup (6, \infty) \times$	(c) $\langle -2, 0 \rangle \cup (0, 1) \times$	(d) $\left(-\frac{1}{2}, \frac{7}{2}\right) \times$	(e) $-\frac{109}{9} \checkmark$	(f) $0 \checkmark$
$\boxed{\sigma}$	$i : \mathbf{W}$	(a) $\left\langle 1, \frac{7}{5} \right\rangle \times$	(b) $(-\infty, -5) \cup (-5, -4) \cup (-1, \infty) \checkmark$	(c) $\langle -5, 2 \rangle \cup (2, 4) \times$	(d) $\left(-\infty, -\frac{1}{5}\right) \cup \left\langle \frac{4}{5}, \infty\right\rangle \checkmark$	(e) $-\frac{185}{27} \checkmark$	(f) $3 \checkmark$
	$ii : \mathbf{I}$	(a) $\left\langle \frac{4}{7}, \frac{6}{7} \right\rangle \times$	(b) $(-\infty, -2) \cup (1, 2) \cup (2, \infty) \checkmark$	(c) $\langle -6, -3 \rangle \cup (-3, 7) \times$	(d) $(-\infty, -2) \cup \left\langle \frac{1}{5}, \infty\right\rangle \times$	(e) $-\frac{227}{27} \times$	(f) $0 \checkmark$
	$iii : \mathbf{L}$	(a) $\left\langle \frac{7}{5}, \frac{11}{5} \right\rangle \checkmark$	(b) $(-\infty, -7) \cup (4, 5) \cup (5, \infty) \checkmark$	(c) $\left\langle -6, \frac{10}{3} \right\rangle \cup \left(\frac{10}{3}, 5\right) \checkmark$	(d) $\left(-\frac{8}{3}, \frac{3}{8}\right) \times$	(e) $-\frac{1713}{64} \times$	(f) $3 \times$
	$iv : \mathbf{D}$	(a) $\left\langle \frac{1}{6}, 1 \right\rangle \checkmark$	(b) $(-\infty, 4) \cup (5, 6) \cup (6, \infty) \times$	(c) $\left\langle -2, -\frac{3}{2} \right\rangle \cup \left(-\frac{3}{2}, 3\right) \times$	(d) $\left\langle \frac{1}{7}, 7 \right\rangle \times$	(e) $-\frac{681}{32} \checkmark$	(f) $0 \checkmark$
$\boxed{\pi}$	$i : \mathbf{V}$	(a) $\left\langle \frac{1}{2}, 2 \right\rangle \checkmark$	(b) $(-\infty, -9) \cup (5, 6) \cup (6, \infty) \checkmark$	(c) $\langle -7, 0 \rangle \cup (0, 6) \checkmark$	(d) $\left\langle -\frac{1}{3}, \frac{4}{5} \right\rangle \checkmark$	(e) $-\frac{17}{27} \times$	(f) $0 \times$
	$ii : \mathbf{I}$	(a) $\langle 6, 11 \rangle \times$	(b) $(-\infty, -4) \cup (-4, -3) \cup (9, \infty) \checkmark$	(c) $\langle -9, 4 \rangle \cup (4, 8) \times$	(d) $\left(-\infty, \frac{1}{6}\right) \cup (3, \infty) \times$	(e) $-\frac{104}{27} \times$	(f) $0 \checkmark$
	$iii : \mathbf{E}$	(a) $\langle 6, 8 \rangle \checkmark$	(b) $(-\infty, -7) \cup (3, 4) \cup (4, \infty) \times$	(c) $\left(-1, -\frac{4}{5}\right) \cup \left(-\frac{4}{5}, 2\right) \times$	(d) $\left\langle -\frac{7}{6}, 5 \right\rangle \times$	(e) $-\frac{380}{27} \checkmark$	(f) $0 \times$
	$iv : \mathbf{W}$	(a) $\left\langle 1, \frac{9}{7} \right\rangle \times$	(b) $(-\infty, -2) \cup (5, 6) \cup (6, \infty) \checkmark$	(c) $\langle -4, -2 \rangle \cup (-2, 5) \times$	(d) $(-\infty, -1) \cup (2, \infty) \checkmark$	(e) $-\frac{15}{16} \checkmark$	(f) $0 \checkmark$

Function of one variable (solutions)

ρ	$i : \mathbf{B}$	(a) $\langle 2, 7 \rangle \checkmark$	(b) $(-\infty, 1) \cup (4, 5) \cup (5, \infty) \checkmark$	(c) $\left(-7, -\frac{49}{8}\right) \cup \left(-\frac{49}{8}, 8\right) \times$	(d) $\left(-\infty, \frac{1}{3}\right) \cup (2, \infty) \times$	(e) $-\frac{219}{32} \times$	(f) $3 \times$
	$ii : \mathbf{A}$	(a) $\langle 2, 3 \rangle \checkmark$	(b) $(-\infty, 1) \cup (1, 2) \cup (3, \infty) \times$	(c) $\left(-2, -\frac{7}{4}\right) \cup \left(-\frac{7}{4}, 3\right) \times$	(d) $\left(-\infty, \frac{5}{7}\right) \cup \left(\frac{4}{5}, \infty\right) \times$	(e) $\frac{19}{2} \times$	(f) $3 \times$
	$iii : \mathbf{B}$	(a) $\left(1, \frac{7}{5}\right) \checkmark$	(b) $(-\infty, -3) \cup (-2, -1) \cup (-1, \infty) \checkmark$	(c) $\left(-2, -\frac{4}{4}\right) \cup \left(-\frac{4}{3}, 3\right) \times$	(d) $\langle -1, 1 \rangle \times$	(e) $\frac{59}{8} \times$	(f) $0 \times$
	$iv : \mathbf{Y}$	(a) $\left\langle \frac{3}{4}, \frac{5}{4} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (3, 4) \cup (4, \infty) \times$	(c) $\left(-1, -\frac{6}{7}\right) \cup \left(-\frac{6}{7}, 2\right) \checkmark$	(d) $\left(-\infty, \frac{1}{3}\right) \cup \left(\frac{1}{2}, \infty\right) \checkmark$	(e) $\frac{127}{8} \checkmark$	(f) $0 \checkmark$
σ	$i : \mathbf{Y}$	(a) $\langle 5, 7 \rangle \checkmark$	(b) $(-\infty, -6) \cup (1, 2) \cup (2, \infty) \times$	(c) $\left(-4, -\frac{16}{5}\right) \cup \left(-\frac{16}{5}, 5\right) \checkmark$	(d) $\left\langle \frac{-6}{7}, \frac{1}{3} \right\rangle \checkmark$	(e) $-\frac{11}{4} \checkmark$	(f) $0 \checkmark$
	$ii : \mathbf{E}$	(a) $\left\langle \frac{4}{3}, \frac{8}{3} \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (-7, -6) \cup (-6, \infty) \times$	(c) $\left(-5, -\frac{35}{8}\right) \cup \left(-\frac{35}{8}, 6\right) \times$	(d) $\langle -4, 2 \rangle \times$	(e) $-\frac{535}{27} \checkmark$	(f) $0 \times$
	$iii : \mathbf{A}$	(a) $\left(2, \frac{7}{2}\right) \checkmark$	(b) $(-\infty, 1) \cup (2, 3) \cup (3, \infty) \times$	(c) $\langle -3, 1 \rangle \cup (1, 2) \times$	(d) $\left\langle \frac{-3}{4}, \frac{-1}{2} \right\rangle \times$	(e) $-\frac{27}{64} \times$	(f) $0 \times$
	$iv : \mathbf{R}$	(a) $\left\langle \frac{4}{5}, \frac{7}{5} \right\rangle \checkmark$	(b) $(-\infty, -9) \cup (-2, -1) \cup (-1, \infty) \checkmark$	(c) $\langle -3, 0 \rangle \cup (0, 2) \checkmark$	(d) $\left\langle -3, \frac{-4}{3} \right\rangle \times$	(e) $-\frac{1371}{64} \checkmark$	(f) $0 \times$
τ	$i : \mathbf{A}$	(a) $\left(2, \frac{8}{3}\right) \checkmark$	(b) $(-\infty, 5) \cup (9, 10) \cup (10, \infty) \times$	(c) $\langle -3, 0 \rangle \cup (0, 4) \times$	(d) $\left\langle \frac{-2}{3}, \frac{9}{2} \right\rangle \times$	(e) $\frac{35}{2} \times$	(f) $0 \times$
	$ii : \mathbf{C}$	(a) $\left\langle \frac{7}{2}, \frac{13}{2} \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (6, 7) \cup (7, \infty) \times$	(c) $\left(-3, -\frac{21}{8}\right) \cup \left(-\frac{21}{8}, 4\right) \times$	(d) $(-\infty, -1) \cup (2, \infty) \times$	(e) $-\frac{671}{27} \times$	(f) $3 \checkmark$
	$iii : \mathbf{I}$	(a) $\left\langle \frac{1}{4}, 1 \right\rangle \times$	(b) $(-\infty, -4) \cup (1, 2) \cup (2, \infty) \checkmark$	(c) $\langle -2, -1 \rangle \cup (-1, 3) \times$	(d) $\left\langle \frac{-1}{4}, \frac{1}{3} \right\rangle \times$	(e) $\frac{472}{27} \times$	(f) $0 \checkmark$
	$iv : \mathbf{D}$	(a) $\left\langle \frac{7}{4}, \frac{5}{2} \right\rangle \checkmark$	(b) $(-\infty, -7) \cup (-5, -4) \cup (-4, \infty) \times$	(c) $\langle -3, 1 \rangle \cup (1, 2) \times$	(d) $\left(-\infty, \frac{-3}{8}\right) \cup \left(\frac{1}{2}, \infty\right) \times$	(e) $\frac{524}{27} \checkmark$	(f) $0 \checkmark$
υ	$i : \mathbf{L}$	(a) $\left\langle \frac{2}{7}, \frac{5}{7} \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (-5, -4) \cup (1, \infty) \checkmark$	(c) $\left\langle -6, \frac{5}{2} \right\rangle \cup \left(\frac{5}{2}, 5\right) \checkmark$	(d) $\left(-\infty, \frac{-4}{3}\right) \cup \left(\frac{8}{3}, \infty\right) \times$	(e) $-\frac{41}{2} \times$	(f) $0 \times$
	$ii : \mathbf{I}$	(a) $\left\langle \frac{3}{-1}, 1 \right\rangle \times$	(b) $(-\infty, 2) \cup (7, 8) \cup (8, \infty) \checkmark$	(c) $\left(-4, -\frac{10}{3}\right) \cup \left(-\frac{10}{3}, 5\right) \times$	(d) $\left(-\infty, \frac{-1}{4}\right) \cup \left\langle \frac{-1}{7}, \infty \right\rangle \times$	(e) $\frac{475}{27} \times$	(f) $3 \checkmark$
	$iii : \mathbf{N}$	(a) $\left\langle \frac{4}{7}, \frac{6}{7} \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (1, 2) \cup (4, \infty) \times$	(c) $\left(-5, -\frac{5}{2}\right) \cup \left(-\frac{5}{2}, 6\right) \checkmark$	(d) $\left\langle \frac{-2}{3}, \frac{1}{6} \right\rangle \times$	(e) $\frac{293}{27} \checkmark$	(f) $0 \checkmark$
	$iv : \mathbf{E}$	(a) $\left\langle \frac{3}{1}, \frac{2}{2} \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (-5, -4) \cup (2, \infty) \times$	(c) $\left\langle -2, \frac{1}{2} \right\rangle \cup \left(\frac{1}{2}, 1\right) \times$	(d) $\left\langle \frac{-5}{3}, \frac{-1}{8} \right\rangle \times$	(e) $\frac{89}{27} \checkmark$	(f) $0 \times$
ϕ	$i : \mathbf{F}$	(a) $\left\langle \frac{4}{3}, 2 \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (2, 3) \cup (3, \infty) \checkmark$	(c) $\langle -3, -2 \rangle \cup (-2, 4) \times$	(d) $\left(-\infty, \frac{-1}{6}\right) \cup \left(\frac{3}{5}, \infty\right) \times$	(e) $-\frac{101}{9} \times$	(f) $0 \checkmark$
	$ii : \mathbf{I}$	(a) $\langle 2, 4 \rangle \times$	(b) $(-\infty, -2) \cup (1, 2) \cup (2, \infty) \checkmark$	(c) $\langle -1, 0 \rangle \cup (0, 2) \times$	(d) $\left\langle -2, \frac{1}{2} \right\rangle \times$	(e) $\frac{235}{9} \times$	(f) $0 \checkmark$
	$iii : \mathbf{V}$	(a) $\left\langle \frac{5}{2}, \frac{7}{2} \right\rangle \checkmark$	(b) $(-\infty, -7) \cup (2, 3) \cup (3, \infty) \checkmark$	(c) $\langle -3, 0 \rangle \cup (0, 4) \checkmark$	(d) $(-\infty, -1) \cup \left(\frac{1}{4}, \infty\right) \checkmark$	(e) $-\frac{1819}{64} \times$	(f) $0 \times$
	$iv : \mathbf{E}$	(a) $\langle 4, 6 \rangle \checkmark$	(b) $(-\infty, -3) \cup (-3, -2) \cup (3, \infty) \times$	(c) $\left(-1, -\frac{7}{8}\right) \cup \left(-\frac{7}{8}, 2\right) \times$	(d) $(-\infty, 1) \cup (8, \infty) \times$	(e) $\frac{19}{16} \checkmark$	(f) $3 \times$
χ	$i : \mathbf{D}$	(a) $\left\langle \frac{7}{6}, \frac{3}{2} \right\rangle \checkmark$	(b) $(-\infty, -1) \cup (3, 4) \cup (4, \infty) \times$	(c) $\langle -3, 0 \rangle \cup (0, 4) \times$	(d) $\left(-\infty, \frac{3}{5}\right) \cup (3, \infty) \times$	(e) $-\frac{56}{9} \checkmark$	(f) $0 \checkmark$
	$ii : \mathbf{A}$	(a) $\langle 3, 9 \rangle \checkmark$	(b) $(-\infty, 2) \cup (2, 3) \cup (4, \infty) \times$	(c) $\left(-1, -\frac{6}{7}\right) \cup \left(-\frac{6}{7}, 2\right) \times$	(d) $(-\infty, -5) \cup \left(\frac{1}{3}, \infty\right) \times$	(e) $\frac{357}{16} \times$	(f) $0 \times$
	$iii : \mathbf{T}$	(a) $\left\langle \frac{5}{8}, \frac{9}{8} \right\rangle \times$	(b) $(-\infty, -10) \cup (-10, -9) \cup (-5, \infty) \checkmark$	(c) $\left(-1, -\frac{3}{4}\right) \cup \left(-\frac{3}{4}, 2\right) \checkmark$	(d) $\left(-\infty, \frac{-6}{5}\right) \cup \left(\frac{-2}{7}, \infty\right) \times$	(e) $-\frac{41}{27} \checkmark$	(f) $0 \checkmark$
	$iv : \mathbf{E}$	(a) $\left\langle \frac{1}{5}, \frac{8}{3} \right\rangle \checkmark$	(b) $(-\infty, -2) \cup (6, 7) \cup (7, \infty) \times$	(c) $\langle -5, 3 \rangle \cup (3, 4) \times$	(d) $\left\langle \frac{-3}{8}, 2 \right\rangle \times$	(e) $\frac{15}{4} \checkmark$	(f) $0 \times$
ψ	$i : \mathbf{C}$	(a) $\left\langle 4, \frac{13}{2} \right\rangle \checkmark$	(b) $(-\infty, -5) \cup (-2, -1) \cup (-1, \infty) \times$	(c) $\left(-5, -\frac{5}{2}\right) \cup \left(-\frac{5}{2}, 6\right) \times$	(d) $\left(-\infty, \frac{-5}{8}\right) \cup \left(\frac{1}{2}, \infty\right) \times$	(e) $-\frac{146}{9} \times$	(f) $0 \checkmark$
	$ii : \mathbf{O}$	(a) $\left\langle \frac{4}{7}, \frac{6}{7} \right\rangle \checkmark$	(b) $(-\infty, -3) \cup (-2, -1) \cup (-1, \infty) \times$	(c) $\left\langle -9, \frac{64}{9} \right\rangle \cup \left(\frac{64}{9}, 8\right) \checkmark$	(d) $\left(-\infty, \frac{7}{9}\right) \cup (1, \infty) \times$	(e) $\frac{146}{9} \checkmark$	(f) $0 \times$
	$iii : \mathbf{L}$	(a) $\left\langle \frac{2}{3}, \frac{7}{6} \right\rangle \checkmark$	(b) $(-\infty, -7) \cup (-7, -6) \cup (3, \infty) \checkmark$	(c) $\left(-2, -\frac{3}{2}\right) \cup \left(-\frac{3}{2}, 3\right) \checkmark$	(d) $\left(-\infty, \frac{1}{5}\right) \cup \left(\frac{1}{2}, \infty\right) \times$	(e) $7 \times$	(f) $0 \times$
	$iv : \mathbf{D}$	(a) $\left\langle \frac{4}{9}, \frac{2}{3} \right\rangle \checkmark$	(b) $(-\infty, -4) \cup (-4, -3) \cup (4, \infty) \times$	(c) $\left\langle -6, \frac{5}{2} \right\rangle \cup \left(\frac{5}{2}, 5\right) \times$	(d) $\left\langle \frac{-5}{3}, \frac{4}{3} \right\rangle \times$	(e) $\frac{19}{2} \checkmark$	(f) $3 \checkmark$
ω	$i : \mathbf{T}$	(a) $\left\langle \frac{2}{3}, 1 \right\rangle \times$	(b) $(-\infty, -6) \cup (-6, -5) \cup (2, \infty) \checkmark$	(c) $\left\langle -3, \frac{8}{5} \right\rangle \cup \left(\frac{8}{5}, 2\right) \checkmark$	(d) $\left\langle -3, \frac{-1}{3} \right\rangle \times$	(e) $-\frac{473}{27} \checkmark$	(f) $3 \checkmark$
	$ii : \mathbf{R}$	(a) $\left\langle \frac{9}{4}, 3 \right\rangle \checkmark$	(b) $(-\infty, 1) \cup (1, 2) \cup (6, \infty) \checkmark$	(c) $\langle -2, -1 \rangle \cup (-1, 3) \checkmark$	(d) $\left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \times$	(e) $\frac{427}{27} \checkmark$	(f) $3 \times$
	$iii : \mathbf{I}$	(a) $\left\langle \frac{3}{8}, \frac{5}{8} \right\rangle \times$	(b) $(-\infty, -7) \cup (-7, -6) \cup (1, \infty) \checkmark$	(c) $\left\langle -3, \frac{12}{7} \right\rangle \cup \left(\frac{12}{7}, 2\right) \times$	(d) $\left\langle -1, \frac{1}{4} \right\rangle \times$	(e) $\frac{320}{27} \times$	(f) $3 \checkmark$
	$iv : \mathbf{P}$	(a) $\left\langle \frac{5}{2}, \frac{7}{2} \right\rangle \checkmark$	(b) $(-\infty, -8) \cup (-8, -7) \cup (-1, \infty) \checkmark$	(c) $\left(-1, -\frac{7}{8}\right) \cup \left(-\frac{7}{8}, 2\right) \checkmark$	(d) $\left\langle \frac{-1}{3}, \frac{-1}{5} \right\rangle \times$	(e) $\frac{3}{16} \times$	(f) $3 \checkmark$